

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12410758
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Ostdiek; David Marion et al.

Three-stream gas turbine engine control

Abstract

A control system and schemes for controlling a three-stream gas turbine engine are disclosed. In one aspect, a three-stream engine is architecturally arranged so as to define a primary bypass flowpath, a secondary bypass flowpath, and a core flowpath that may each output propulsive thrust. The three-stream engine includes one or more effectors that can be controlled to adjust a thrust contribution provided by the secondary bypass flowpath to the net propulsive thrust as well as a thermal contribution provided by the secondary bypass flowpath to an associated thermal management system. Competing demands, limits, and priorities can be considered in controlling the effector. In some embodiments, a secondary effector can be ganged or controlled in conjunction with the effector to assist with adjustment of the contributions provided by the secondary bypass flowpath.

Inventors: Ostdiek; David Marion (Liberty Township, OH), Spruill; Jeffrey S. (Hillsboro, OH), Cafaro; Stefan Joseph (Chapel Hill, NC), Hiett; David Alexander (Cincinnati, OH)

Applicant: General Electric Company (Schenectady, NY)

Family ID: 84367561

Assignee: General Electric Company (Schenectady, NY)

Appl. No.: 17/571642

Filed: January 10, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230220815 A1	Jul. 13, 2023

Publication Classification

Int. Cl.: F02C9/20 (20060101); F02K3/02 (20060101); F02K3/075 (20060101); F02K3/077 (20060101)

U.S. Cl.:

CPC F02C9/20 (20130101); F02K3/02 (20130101); F02K3/075 (20130101); F02K3/077 (20130101); F05D2270/303 (20130101)

Field of Classification Search

CPC: F02K (3/02); F02K (3/075); F02K (3/077); F02C (9/20)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4064692	12/1976	Johnson et al.	N/A	N/A
4446696	12/1983	Sargisson	60/226.3	F02K 3/077
4534526	12/1984	Metzger et al.	N/A	N/A
4917336	12/1989	Jacobs et al.	N/A	N/A
4934825	12/1989	Martin	N/A	N/A
4976102	12/1989	Taylor	N/A	N/A
5148402	12/1991	Magliozzi et al.	N/A	N/A
5156353	12/1991	Gliebe et al.	N/A	N/A
5190441	12/1992	Murphy et al.	N/A	N/A
5197855	12/1992	Magliozzi et al.	N/A	N/A
5261227	12/1992	Giffin, III	60/226.3	F01D 17/162
5789678	12/1997	Pla	N/A	N/A
6341747	12/2001	Schmidt et al.	N/A	N/A
6735955	12/2003	Mannarino	60/794	F04D 27/0261
7578132	12/2008	Webster	N/A	N/A
7762084	12/2009	Martis	60/785	F01D 17/162
8011613	12/2010	Belleville	N/A	N/A
8083482	12/2010	Serven et al.	N/A	N/A
8087607	12/2011	Moore et al.	N/A	N/A
8186617	12/2011	Llamas Sandin	N/A	N/A
8196860	12/2011	Gall et al.	N/A	N/A
8210798	12/2011	Stern	N/A	N/A
8220586	12/2011	Todorovic	N/A	N/A
8240609	12/2011	Parazzoli et al.	N/A	N/A
8350398	12/2012	Butt	N/A	N/A
8352099	12/2012	Eggold	701/4	B64C 19/00
8550397	12/2012	Verde Preckler et al.	N/A	N/A
8690106	12/2013	Reissig	N/A	N/A
8967967	12/2014	Stretton et al.	N/A	N/A
9057329	12/2014	Weir et al.	N/A	N/A
9102397	12/2014	Wood	N/A	N/A

9527578	12/2015	Towkan	N/A	N/A
9540094	12/2016	Negulescu et al.	N/A	N/A
9677501	12/2016	Pierluissi et al.	N/A	N/A
9725155	12/2016	Miller et al.	N/A	N/A
9745051	12/2016	Tantot et al.	N/A	N/A
9758254	12/2016	Moore et al.	N/A	N/A
9845768	12/2016	Pesyna et al.	N/A	N/A
9856745	12/2017	Ali	N/A	N/A
9890657	12/2017	Ali	N/A	N/A
9920710	12/2017	Dawson et al.	N/A	N/A
10118710	12/2017	Derrez et al.	N/A	N/A
10131441	12/2017	Edwards et al.	N/A	N/A
10190506	12/2018	Ruberte Sanchez	N/A	N/A
10202865	12/2018	Breeze-Stringfellow et al.	N/A	N/A
10364750	12/2018	Rambo	N/A	N/A
10370086	12/2018	Vion et al.	N/A	N/A
10371090	12/2018	Lovett et al.	N/A	N/A
10378452	12/2018	Barmichev et al.	N/A	N/A
10399664	12/2018	Bowden et al.	N/A	N/A
10400710	12/2018	Dawson et al.	N/A	N/A
10414486	12/2018	Wood et al.	N/A	N/A
10414497	12/2018	Velez De Mendizabal Alonso et al.	N/A	N/A
10669881	12/2019	Breeze-Stringfellow et al.	N/A	N/A
10787996	12/2019	Kupratis et al.	N/A	N/A
2007/0000232	12/2006	Powell et al.	N/A	N/A
2008/0245925	12/2007	Udall	N/A	N/A
2011/0115227	12/2010	Shafer	290/52	F02K 3/115
2011/0150627	12/2010	Baughman	N/A	N/A
2011/0167792	12/2010	Johnson et al.	N/A	N/A
2012/0195739	12/2011	Kingan	N/A	N/A
2013/0115083	12/2012	Vuillemin	N/A	N/A
2014/0248116	12/2013	Ali	N/A	N/A
2014/0263737	12/2013	Pierluissi et al.	N/A	N/A
2015/0291276	12/2014	Zatorski et al.	N/A	N/A
2015/0344127	12/2014	Wood et al.	N/A	N/A
2016/0010487	12/2015	Breeze-Stringfellow et al.	N/A	N/A
2016/0333729	12/2015	Miller et al.	N/A	N/A
2016/0333734	12/2015	Bowden et al.	N/A	N/A
2016/0368592	12/2015	Szymandera	N/A	N/A
2017/0002688	12/2016	Beutin et al.	N/A	N/A
2017/0159571	12/2016	Sidelkovskiy	N/A	N/A
2017/0225773	12/2016	Wood et al.	N/A	N/A
2018/0065727	12/2017	Gruber et al.	N/A	N/A
2018/0079492	12/2017	Seidel et al.	N/A	N/A
2018/0127084	12/2017	Tajan et al.	N/A	N/A

2018/0354634	12/2017	Jodet et al.	N/A	N/A
2019/0128189	12/2018	Rambo	N/A	N/A
2020/0088109	12/2019	Tang	N/A	F02C 9/20
2020/0332726	12/2019	Kehoe et al.	N/A	N/A
2021/0139156	12/2020	Hamel	N/A	F02C 9/28

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2971723	12/2015	EP	N/A
WO2011038216	12/2010	WO	N/A

OTHER PUBLICATIONS

Carney et al., Weight Assessment for Fuselage Shielding on Aircraft With Open-Rotor Engines and Composite Blade Loss, NASA/TM-2013-216582, 2013, 33 Pages. cited by applicant

Gazzaniga et al., Wind Tunnel Performance Results of Swirl Recovery Vanes as Tested with an Advanced High-Speed Propeller, AIAA-92-3770, 28th Joint Propulsion Conference Exhibit, Nashville TN, Jul. 6-8, 1992, 42 Pages. cited by applicant

Open Rotor Engine and Installation, European Aviation Safety Agency, Notice of Proposed Amendment 2015-22, RMT.0384 (MDM.092) Dec. 21, 2015., 96 Pages. cited by applicant

Primary Examiner: Meade; Lorne E

Attorney, Agent or Firm: Smith Gambrell & Russell LLP

Background/Summary

FIELD

(1) The present subject matter relates generally to gas turbine engines, and more particularly to control of gas turbine engines.

BACKGROUND

(2) Some gas turbine engines for aircraft have or are coupled with a thermal management system that manage the engine and/or aircraft heat loads. Such gas turbine engines are also typically responsible for producing the thrust needed to propel the aircraft. Control systems and schemes for controlling a thrust contribution of a gas turbine engine as well as a thermal contribution to its associated thermal management system would be a welcome addition to the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present subject matter, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

(2) FIG. 1 provides a schematic cross-sectional view of a three-stream gas turbine engine according to various embodiments of the present disclosure;

(3) FIG. 2 provides a close-up, schematic cross-sectional view of a forward portion of the three-stream engine of FIG. 1;

(4) FIGS. 3A, 3B, and 3C provide schematic cross-sectional views of various alternative

embodiments of three-stream gas turbine engines according to yet other various embodiments of the present disclosure;

(5) FIG. 4 provides a system diagram of a control system for an aircraft according to various embodiments of the present disclosure;

(6) FIG. 5 provides a graph depicting inlet guide vane position scheduled as a function of a power level of the three-stream engine of FIGS. 1 and 2;

(7) FIG. 6 provides a graph depicting inlet guide vane position as a function of time in response to an increased thermal demand according to one example embodiment of the present disclosure;

(8) FIG. 7 provides a graph depicting inlet guide vane position as a function of time in response to an increased thrust demand according to one example embodiment of the present disclosure;

(9) FIG. 8 provides a graph depicting inlet guide vane position as a function of time in response to an increased thermal demand and an increased thrust demand according to one example embodiment of the present disclosure;

(10) FIG. 9 provides a graph depicting variable nozzle position scheduled as a function of a power level of the three-stream engine of FIGS. 1 and 2;

(11) FIG. 10 provides a graph depicting various speed lines scheduled by pitch as a function of a corrected thrust of the three-stream engine of FIGS. 1 and 2;

(12) FIG. 11 provides a graph depicting a power ratio scheduled as a function of a power level of the three-stream engine of FIGS. 1 and 2;

(13) FIG. 12 provides a graph depicting booster variable stator vane position scheduled as a function of a power level of the three-stream engine of FIGS. 1 and 2;

(14) FIG. 13 provides a graph depicting variable bleed valve scheduled as a function of a power level of the three-stream engine of FIGS. 1 and 2;

(15) FIG. 14 provides a flow diagram for a method of operating a three-stream engine according to various embodiments of the present disclosure; and

(16) FIG. 15 provides a block diagram of a computing system for implementing one or more aspects of the present disclosure according to example embodiments of the present disclosure.

DETAILED DESCRIPTION

(17) Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

(18) As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or relative importance of the individual components.

(19) The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

(20) The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

(21) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

(22) The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers only A, only B, only C, or any combination of A, B, and C.

(23) Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an

instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 2, 5, 10, or 20 percent margin.

(24) Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

(25) The term “turbomachine” or “turbomachinery” refers to a machine including one or more compressors, a heat generating section (e.g., a combustion section), and one or more turbines that together generate a torque output.

(26) The term “gas turbine engine” refers to an engine having a turbomachine as all or a portion of its power source. Example gas turbine engines include turbofan engines, turboprop engines, turbojet engines, turboshaft engines, etc., as well as hybrid-electric versions of one or more of these engines.

(27) The term “combustion section” refers to any heat addition system for a turbomachine. For example, the term combustion section may refer to a section including one or more of a deflagrative combustion assembly, a rotating detonation combustion assembly, a pulse detonation combustion assembly, or other appropriate heat addition assembly. In certain example embodiments, the combustion section may include an annular combustor, a can combustor, a cannular combustor, a trapped vortex combustor (TVC), or other appropriate combustion system, or combinations thereof.

(28) The terms “low” and “high”, or their respective comparative degrees (e.g., -er, where applicable), when used with a compressor, a turbine, a shaft, or spool components, etc. each refer to relative speeds within an engine unless otherwise specified. For example, a “low turbine” or “low speed turbine” defines a component configured to operate at a rotational speed, such as a maximum allowable rotational speed, lower than a “high turbine” or “high speed turbine” at the engine.

(29) Aspects of the present disclosure are directed to control systems and schemes for controlling a three-stream gas turbine engine. In one aspect, a three-stream engine is provided that is architecturally arranged so as to produce three distinct streams of propulsive thrust that collectively provide the net propulsive thrust of the engine. The three-stream gas turbine engine defines a primary bypass flowpath, a secondary bypass flowpath, and a core flowpath that may each output propulsive thrust. The three-stream engine can include one or more effectors that can be controlled to adjust a thrust contribution provided by the secondary bypass flowpath to the net propulsive thrust as well as a thermal contribution provided by the secondary bypass flowpath to an associated thermal management system. Competing demands, limits, and priorities can be considered in controlling the effector. In some embodiments, a secondary effector can be ganged or controlled in conjunction with the effector to assist with adjustment of the contributions provided by the secondary bypass flowpath.

(30) Three-stream engines architecturally arranged and operable to implement one or more of the disclosed control schemes may have certain advantages and benefits. For instance, such control schemes may allow for optimized or otherwise improved performance and constraint handling of three-stream engines. Priority of the thrust capability and/or the cooling capability of such three-stream engines can be adjusted for optimized or otherwise improved performance whilst maintaining operability and safe operation. Specifically, such control schemes may take into consideration compressor operating lines, thermal constraints or demands, fan speed or propeller torque, limits associated with variable geometry components, as well as the controllability of the aircraft or vehicle to which the three-stream engine is mounted. Operable, rapid transient capability may be achieved by such three-stream engines. Other benefits and advantages may be realized as well.

(31) Turning now to the drawings, FIG. 1 provides a schematic cross-sectional view of a gas

turbine engine according to one example embodiment of the present disclosure. Particularly, FIG. 1 provides an aviation three-stream gas turbine engine herein referred to as “three-stream engine **100**”. The three-stream engine **100** of FIG. 1 can be mounted to an aerial vehicle, such as a fixed-wing aircraft, and can produce thrust for propulsion of the aerial vehicle. The three-stream engine **100** is a “three-stream engine” in that its architecture provides three distinct streams of thrust-producing airflow during operation.

(32) For reference, the three-stream engine **100** defines an axial direction A, a radial direction R, and a circumferential direction C. Moreover, the three-stream engine **100** defines an axial centerline or longitudinal axis **112** that extends along the axial direction A. In general, the axial direction A extends parallel to the longitudinal axis **112**, the radial direction R extends outward from and inward to the longitudinal axis **112** in a direction orthogonal to the axial direction A, and the circumferential direction extends three hundred sixty degrees (360°) around the longitudinal axis **112**. The three-stream engine **100** extends between a forward end **114** and an aft end **116**, e.g., along the axial direction A.

(33) The three-stream engine **100** includes a core engine **118** and a fan section **150** positioned upstream thereof. Generally, the core engine **118** includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. Particularly, as shown in FIG. 1, the core engine **118** includes an engine core **120** and a core cowl **122** that annularly surrounds at least a portion of the engine core **120**. The engine core **120** and core cowl **122** define an annular core inlet **124**. The core cowl **122** further encloses and supports a booster or low pressure compressor **126** for pressurizing the air that enters the core engine **118** through the core inlet **124**. A high pressure, multi-stage, axial-flow compressor **128** receives pressurized air from the LP compressor **126** and further increases the pressure of the air. The pressurized air stream flows downstream to a combustor **130** where fuel is injected into the pressurized air stream and ignited to raise the temperature and energy level of the pressurized air.

(34) The high energy combustion products flow from the combustor **130** downstream to a high pressure turbine **132**. The HP turbine **132** drives the HP compressor **128** through a first shaft or high pressure shaft **136**. In this regard, the HP turbine **132** is drivingly coupled with the HP compressor **128**. The high energy combustion products then flow to a low pressure turbine **134**. The LP turbine **134** drives the LP compressor **126**, components of the fan section **150**, and optionally, an electric machine **200** through a second shaft or low pressure shaft **138**. In this regard, the LP turbine **134** is drivingly coupled with the LP compressor **126**, components of the fan section **150**, and the electric machine **200**. The LP shaft **138** is coaxial with the HP shaft **136** in this example embodiment. After driving each of the turbines **132**, **134**, the combustion products exit the core engine **118** through a core exhaust nozzle **140** to produce propulsive thrust. Accordingly, the core engine **118** defines a core flowpath **142** that extends between the core inlet **124** and the core exhaust nozzle **140**. The core flowpath **142** is an annular flowpath positioned generally inward of the core cowl **122** along the radial direction R.

(35) The fan section **150** includes a primary fan **152**. For the depicted embodiment of FIG. 1, the primary fan **152** is an open rotor or unducted primary fan. However, in other embodiments, the primary fan **152** may be a ducted fan. For instance, in FIG. 3A, the primary fan **152** is shown ducted by a fan casing **157** or nacelle circumferentially surrounding the primary fan **152**. Returning to FIG. 1, the primary fan **152** includes an array of fan blades **154** (only one shown in FIG. 1). The fan blades **154** are rotatable, e.g., about the longitudinal axis **112**. As noted above, the primary fan **152** is drivingly coupled with the LP turbine **134** via the LP shaft **138**. In some embodiments, the primary fan **152** can be directly coupled with the LP shaft **138**, e.g., in a direct-drive configuration. In other embodiments, as shown in FIG. 1, the primary fan **152** can be mechanically coupled with the LP shaft **138** via a speed reduction gearbox **155**, e.g., in an indirect-drive or geared-drive configuration.

(36) Moreover, the fan blades **154** can be arranged in equal spacing around the longitudinal axis

112. Each blade **154** has a root and a tip and a span defined therebetween. Each blade **154** defines a central blade axis **156**. For this embodiment, each blade **154** of the primary fan **152** is rotatable about their respective central blade axes **156**, e.g., in unison with one another. One or more actuators **158** can be controlled to pitch the fan blades **154** about their respective central blade axes **156**. As will be explained further herein, the fan blades **154** can be pitched or rotated about their respective central blade axes **156** to effect or control the airflow through the secondary bypass flowpath **172**. In this way, the fan blades **154** can be modulated to optimize or otherwise improve the thrust contribution of the secondary bypass flowpath **172** or to optimize or otherwise improve the thermal management contribution provided by the secondary bypass flowpath **172**, or stated another way, its interaction with a thermal management system.

(37) The fan section **150** further includes a fan guide vane array **160** that includes fan guide vanes **162** (only one shown in FIG. 1) disposed around the longitudinal axis **112**. For this embodiment, the fan guide vanes **162** are not rotatable about the longitudinal axis **112**. Each fan guide vane **162** has a root and a tip and a span defined therebetween. The fan guide vanes **162** may be unshrouded as shown in FIG. 1 or may be shrouded, e.g., by an annular shroud spaced outward from the tips of the fan guide vanes **162** along the radial direction R. Each fan guide vane **162** defines a central blade axis **164**. For this embodiment, each fan guide vane **162** of the fan guide vane array **160** is rotatable about their respective central blades axes **164**, e.g., in unison with one another. One or more actuators **166** can be controlled to pitch the fan guide vane **162** about their respective central blades axes **164**. However, in other embodiments, each fan guide vane **162** may be fixed or unable to be pitched. The fan guide vanes **162** are mounted to an outer cowl **170**.

(38) The outer cowl **170** annularly encases at least a portion of the core cowl **122** and is generally positioned outward of the core cowl **122** along the radial direction R. Particularly, a downstream section of the outer cowl **170** extends over a forward portion of the core cowl **122** to define a secondary bypass flowpath **172**. Incoming air may enter through the secondary bypass flowpath **172** through a secondary bypass inlet **176** and may exit through a secondary bypass outlet **178** to produce propulsive thrust. The secondary bypass flowpath **172** is an annular flowpath generally defined outward of the core flowpath **142** along the radial direction R. The outer cowl **170** and the core cowl **122** are connected together and supported by a plurality of substantially radially-extending, circumferentially-spaced struts **174** (only one shown in FIG. 1). The struts **174** may each be aerodynamically contoured to direct air flowing thereby. Other struts in addition to struts **174** may be used to connect and support the outer cowl **170** and/or core cowl **122**.

(39) Referring now to FIGS. 1 and 2, FIG. 2 provides a close-up, schematic cross-sectional view of a forward portion of the three-stream engine **100**. As depicted, one or more heat exchangers **230** can be incorporated into or within the secondary bypass flowpath **172**. The one or more heat exchangers **230** may be configured to remove or accept heat from various fluids used in engine operation (such as an air-cooled oil cooler (ACOC), cooled cooling air (CCA), etc.). The heat exchangers **230** may take advantage of the integration into the secondary bypass flowpath **172** with reduced performance penalties (such as fuel efficiency and thrust) compared with traditional engine architectures due to not impacting the primary source of thrust which is, in this case, the unducted fan stream labeled as S1 in FIG. 1. The heat exchangers **230** may cool fluids such as gearbox oil, engine sump oil, thermal transport fluids, such as supercritical fluids or commercially available single-phase or two-phase fluids (supercritical CO2, EGV, Syltherm 800, liquid metals, etc.), engine bleed air, etc. The heat exchangers **230** may also be made up of different segments or passages that cool different working fluids, such as an ACOC paired with a fuel cooler. One or more of the heat exchangers **230** can have an associated temperature sensor operable to sense the temperature at the heat exchanger **230**. Various third stream effectors can be controlled based on such temperature readings.

(40) The heat exchangers **230** may be incorporated into a thermal management system that provides for thermal transport via a heat exchange fluid flowing through a network or thermal bus

to remove heat from a source and transport it to a heat exchanger, such as one or more of heat exchangers **230**. One such system is described in commonly-assigned, issued U.S. Pat. No. 10,260,419, which is incorporated herein by reference.

(41) The three-stream engine **100** also defines an inlet flowpath **180**. The inlet flowpath **180** extends between an engine inlet **182** and the core inlet **124**/secondary bypass inlet **176**. The engine inlet **182** is defined generally at the forward end of the outer cowl **170** and is positioned between the primary fan **152** and the array of fan guide vanes **160** along the axial direction A. The inlet flowpath **180** is an annular flowpath that is defined inward of the outer cowl **170** along the radial direction R. Air flowing downstream along the inlet flowpath **180** is split, not necessarily evenly, into the core flowpath **142** and the secondary bypass flowpath **172** by a nose of a splitter **144** of the core cowl **122**. The inlet flowpath **180** is wider than the core flowpath **142** along the radial direction R. The inlet flowpath **180** is also wider than the secondary bypass flowpath **172** along the radial direction R.

(42) As depicted, the fan section **150** also includes a mid-fan **190**. The mid-fan **190** includes an array of mid-fan blades **192** (only one shown in FIG. 1). The mid-fan blades **192** are rotatable, e.g., about the longitudinal axis **112**. The mid-fan **190** is drivingly coupled with the LP turbine **134** via the LP shaft **138**. The mid-fan blades **192** can be arranged in equal circumferential spacing around the longitudinal axis **112**. As shown in FIG. 2, each mid-fan blade **192** has a root **194** and a tip **196** and a span defined therebetween. Moreover, each mid-fan blade **192** has a leading edge **198** and a trailing edge **199**. The mid-fan blades **192** are annularly surrounded or ducted by the outer cowl **170**. In this regard, the mid-fan **190** is positioned inward of the outer cowl **170** along the radial direction R. Moreover, for this example embodiment, the mid-fan **190** is positioned within the inlet flowpath **180** upstream of both the core flowpath **142** and the secondary bypass flowpath **172**.

(43) Air flowing through the inlet flowpath **180** flows across the mid-fan blades **192** and is accelerated downstream thereof, particularly at the tips **196** of the mid-fan blades **192**. At least a portion of the air accelerated by the mid-fan blades **192** flows into the secondary bypass flowpath **172** and is ultimately exhausted through the secondary bypass outlet **178** to produce propulsive thrust. Also, at least a portion of the air accelerated by the mid-fan blades **192** flows into the core flowpath **142** and is ultimately exhausted through the core exhaust nozzle **140** to produce propulsive thrust. Generally, the mid-fan **190** is a compression device positioned downstream of the engine inlet **182**. The mid-fan **190** is operable to accelerate air into the secondary bypass flowpath **172** or secondary bypass passage.

(44) As further shown in FIGS. 1 and 2, an array of variable inlet guide vanes **240** is positioned upstream of the mid-fan **190**. Particularly, the variable inlet guide vanes **240** are positioned immediately upstream of the mid-fan **190** and downstream of the engine inlet **182**. Moreover, the variable inlet guide vanes **240** can be arranged in equal spacing around the longitudinal axis **112**. Each variable inlet guide vane **240** defines a central vane axis **242**. For this embodiment, the variable inlet guide vanes **240** are rotatable about their respective central vane axes **242**, e.g., in unison with one another. One or more actuators **244** can be controlled to pitch the variable inlet guide vanes **240** about their respective central vane axes **242**. As will be explained further herein, the variable inlet guide vanes **240** can be pitched or rotated about their respective central vane axes **242** to effect or control the airflow through the secondary bypass flowpath **172**. In this way, the variable inlet guide vanes **240** can be modulated to optimize or otherwise improve the thrust contribution of the secondary bypass flowpath **172** or to optimize or otherwise improve the thermal management contribution provided by the secondary bypass flowpath **172**, or rather, interaction between the secondary bypass flowpath **172** and a thermal management system.

(45) In some embodiments, optionally, the three-stream engine **100** can include a variable nozzle **250** (see FIG. 2) incorporated into or positioned along the secondary bypass flowpath **172**. For this embodiment, the variable nozzle **250** is positioned at or immediately upstream of the secondary bypass outlet **178**. The variable nozzle **250** can be a slidable, moveable, and/or translatable plug.

For instance, the variable nozzle **250** can be moved between a first or open position and a second or closed position. In FIG. 2, the variable nozzle **250** is shown in the open position via solid lines and in the closed position via the phantom lines. In some embodiments, the variable nozzle **250** can be moved to a position between the open and closed positions, i.e., to an intermediate position. The variable nozzle **250** can be moved by an actuator **252** to vary the exit area through the secondary bypass outlet **178**. The variable nozzle **250** can be an annular, symmetrical device that can be controlled to regulate the airflow through the secondary bypass flowpath **172**. Other suitable nozzle designs may be employed as well, including those incorporating thrust reversing functionality. As will be explained further herein, the variable nozzle **250** can be moved to effect or control the airflow through the secondary bypass flowpath **172**. In this way, the variable nozzle **250** can be modulated to optimize or otherwise improve the thrust contribution of the secondary bypass flowpath **172** or to optimize or otherwise improve the thermal management contribution provided by the secondary bypass flowpath **172**, or stated differently, its interaction with a thermal management system.

(46) The three-stream engine **100** can include other variable geometries as well. For instance, the booster or LP compressor **126** can include one or more stages of compressor rotor blades and compressor stator vanes, wherein at least one stage of compressor stator vanes are variable stator vanes. For instance, as shown in FIGS. 1 and 2, the LP compressor **126** includes at least one stage of compressor stator vanes, which are booster inlet guide vanes **260** in this example. The booster inlet guide vanes **260** are positioned downstream of the mid-fan **190** and are positioned within the core flowpath **142**. The booster inlet guide vanes **260** can be arranged in equal spacing around the longitudinal axis **112**. Each booster inlet guide vane **260** defines a central vane axis **262**. For this embodiment, each booster inlet guide vane **260** is rotatable about their respective central vane axes **262**, e.g., in unison with one another. One or more actuators **264** can be controlled to pitch the booster inlet guide vanes **260** about their respective central vane axes **262**. As will be explained further herein, the booster inlet guide vanes **260** can be pitched or rotated about their respective central vane axes **262** to assist with controlling one or more properties of the airflow through the secondary bypass flowpath **172**.

(47) In addition, as shown in FIG. 2, the three-stream engine **100** can include a variable bleed valve **270** positioned downstream of the LP compressor **126** and upstream of the HP compressor **128**. By modulating the variable bleed valve **270**, air may be bled from the core flowpath **142**. Air can be bled from the core flowpath **142** so that debris can be extracted or expelled from the core flowpath **142**, and to change the operating line of the compressor systems relative to their respective stall lines, among other reasons. The three-stream engine **100** defines at least one of an offboard bleed duct **272** and a secondary bleed duct **274**. The offboard bleed duct **272** provides fluid communication between the core flowpath **142** and offboard the three-stream engine **100**. In this regard, air bled from the core flowpath **142** via the offboard bleed duct **272** is expelled offboard the three-stream engine **100**, e.g., into a primary propulsion stream or first thrust stream **S1** (FIG. 1). The secondary bleed duct **274** provides fluid communication between the core flowpath **142** and the secondary bypass flowpath **172**. In this way, air bled from the core flowpath **142** via the secondary bleed duct **274** is expelled into the secondary bypass flowpath **172**. As will be explained further herein, the variable bleed valve **270** can be modulated to assist with controlling one or more properties of the airflow through the secondary bypass flowpath **172**.

(48) Embodiments of the three-stream engine **100** provided herein generate an increased unducted rotor efficiency at and above a threshold power loading (i.e., power/area of rotor airfoil). In certain embodiments, the threshold power loading is 25 horsepower per ft.² or greater at cruise altitude. In particular embodiments of the engine, structures and methods provided herein generate power loading between 25 horsepower/ft.² and 100 horsepower/ft.² at cruise altitude. Cruise altitude is generally an altitude at which an aircraft levels after climb and prior to descending to an approach flight phase. In various embodiments, the engine is applied to a vehicle

with a cruise altitude up to approximately 65,000 ft. In certain embodiments, cruise altitude is between approximately 28,000 ft and approximately 45,000 ft. In still certain embodiments, cruise altitude is expressed in flight levels based on a standard air pressure at sea level, in which a cruise flight condition is between FL280 and FL650. In another embodiment, cruise flight condition is between FL280 and FL450. In still certain embodiments, cruise altitude is defined based at least on a barometric pressure, in which cruise altitude is between approximately 4.85 psia and approximately 0.82 psia based on a sea level pressure of approximately 14.70 psia and sea level temperature at approximately 59 degree Fahrenheit. In another embodiment, cruise altitude is between approximately 4.85 psia and approximately 2.14 psia. It should be appreciated that in certain embodiments, the ranges of cruise altitude defined by pressure may be adjusted based on a different reference sea level pressure and/or sea level temperature.

(49) As such, it will be appreciated that an engine of such a configuration is configured to generate between about 25,000 and 35,000 pounds of thrust during operation at a rated speed. It will be appreciated, however, that the inventive aspects of the present disclosure are applicable to engines operable to produce between about 2,000 and 130,000 pounds of thrust during operation at a rated speed. Moreover, the inventive aspects of the present disclosure are applicable to engines operable to produce between about 1,000 and 130,000 pounds of thrust during operation.

(50) For the exemplary embodiment of FIG. 1, the primary fan **152** includes twelve (12) fan blades **154**. From a loading standpoint, such a blade count may allow a span of each blade **154** to be reduced such that the overall diameter of the primary fan **152** may also be reduced (e.g., to about twelve feet in the exemplary embodiment). That said, in other embodiments, the primary fan **152** may have any suitable blade count and any suitable diameter. In certain suitable embodiments, the primary fan **152** includes at least eight (8) blades **154**. In another suitable embodiment, the primary fan **152** may have at least twelve (12) blades **154**. In yet another suitable embodiment, the primary fan **152** may have at least fifteen (15) blades **154**. In yet another suitable embodiment, the primary fan **152** may have at least eighteen (18) blades **154**. In one or more of these embodiments, the primary fan **152** includes twenty-six (26) or fewer blades **154**, such as twenty (20) or fewer blades **154**. Further, in certain exemplary embodiments, the primary fan **152** may define a diameter of at least 10 feet, such as at least 11 feet, such as at least 12 feet, such as at least 13 feet, such as at least 15 feet, such as at least 17 feet, such as up to 28 feet, such as up to 26 feet, such as up to 24 feet, such as up to 16 feet.

(51) In various embodiments, it will be appreciated that the three-stream engine **100** includes a ratio of a quantity of vanes **162** to a quantity of blades **154** that could be less than, equal to, or greater than 1:1. For example, in certain embodiments, the three-stream engine **100** may include a ratio of a quantity of vanes **162** to a quantity of blades **154** between 1:2 and 5:2. The ratio may be tuned based on a variety of factors including a size of the vanes **162** to ensure a desired amount of swirl is removed for an airflow from the primary fan **152**.

(52) It should be appreciated that various embodiments of the single unducted rotor engine depicted and described herein may allow for normal subsonic aircraft cruise altitude operation at or above Mach 0.5. In certain embodiments, the engine **100** allows for normal aircraft operation between Mach 0.55 and Mach 0.85 at cruise altitude. In certain embodiments, the engine **100** allows for fan tip speeds (i.e., the tip speeds of the fan blades **154**) at or less than 750 feet per second (fps). As will further be appreciated from the description herein, a loading of the fan blades **154** of the primary fan **152** or rotor assembly may facilitate such flight speeds.

(53) In addition, the three-stream engine **100** can be arranged to define a primary fan radius to mid-fan radius ratio. The primary fan radius to mid-fan radius ratio is defined as: Primary Fan Radius/Mid-Fan Radius. The Primary Fan Radius is measured as a radial length or radius spanning along the radial direction R between the longitudinal axis **112** and a leading edge tip of one of the fan blades **154** of the primary fan **152**. Particularly, as shown best in FIG. 1, the Primary Fan Radius is measured as Radius R1, which spans along the radial direction R between the

longitudinal axis **112** and a leading edge tip of one of the primary fan blades **154**. The Mid-Fan Radius is measured as a radial length or radius spanning along the radial direction R between the longitudinal axis **112** and a leading edge tip of one of the mid-fan blades **192** of the mid-fan **190**. Particularly, as shown in FIG. 2, the Mid-Fan Radius is measured as Radius R2, which spans along the radial direction R between the longitudinal axis **112** and a leading edge tip of one of the mid-fan blades **192**.

(54) In some embodiments, the three-stream engine **100** defines the primary fan radius to mid-fan radius ratio as being equal to or greater than 2.0 and less than or equal to 6.5. Particularly, in some embodiments, the three-stream engine **100** defines the primary fan radius to mid-fan radius ratio as being at least about 2.0. In other embodiments, the three-stream engine **100** defines the primary fan radius to mid-fan radius ratio as being at least about 2.5. In yet other embodiments, the three-stream engine **100** defines the primary fan radius to mid-fan radius ratio as being at least about 3.0. For instance, in FIG. 1, the primary fan radius to mid-fan radius ratio is slightly greater than 3.0. In some further embodiments, the three-stream engine **100** defines the primary fan radius to mid-fan radius ratio as being at least about 4.0. In yet other embodiments, the three-stream engine **100** defines the primary fan radius to mid-fan radius ratio as being at least about 6.0. In some other embodiments, the three-stream engine **100** defines the primary fan radius to mid-fan radius ratio as being about 6.5. For the embodiments having the stated lower bounds of the primary fan radius to mid-fan radius ratio mentioned in this paragraph, unless stated otherwise, the upper bound of these noted ratios may be up to 6.5. The inventors of the present disclosure have recognized that three-stream engines having a primary fan and mid-fan arranged according to the noted ranges/ratios advantageously balance aerodynamic performance and engine efficiency with mechanical constraints of the primary fan **152** and mid-fan **190**.

(55) With reference to FIG. 1, operation of the three-stream engine **100** may be summarized in the following exemplary manner. During operation, an initial or incoming airflow passes through the fan blades **154** of the primary fan **152** and splits into a first airflow and a second airflow. The first airflow bypasses the engine inlet **182** and flows generally along the axial direction A outward of the outer cowl **170** along the radial direction R. The first airflow accelerated by the primary fan blades **154** passes through the fan guide vanes **162** and continues along a primary bypass flowpath **188** defined by the three-stream engine downstream to produce a primary propulsion stream or first thrust stream S1. The vast majority of the net thrust produced by the three-stream engine **100** is produced by the first thrust stream S1. The second airflow enters the inlet flowpath **180** through annular engine inlet **182**.

(56) The second airflow flowing downstream through the inlet flowpath **180** flows through the mid-fan blades **192** of the mid-fan **190** and is consequently compressed. The second airflow flowing downstream of the mid-fan **190** is split by the splitter **144** located at the forward end of the core cowl **122**. Particularly, a portion of the second airflow flowing downstream of the mid-fan **190** flows into the core flowpath **142** through the core inlet **124**. The portion of the second airflow that flows into the core flowpath **142** is progressively compressed by the LP compressor **126** and HP compressor **128** and is ultimately discharged into the combustion section. The discharged pressurized air stream flows downstream to the combustor **130** where fuel is introduced to generate combustion gases or products.

(57) More particularly, the combustor **130** defines an annular combustion chamber that is generally coaxial with the longitudinal centerline axis **112**. The combustor **130** receives an annular stream of pressurized air from the HP compressor **128** via a pressure compressor discharge outlet. A portion of this compressor discharge air flows into a mixer (not shown). Fuel is injected by a fuel nozzle to mix with the air thereby forming a fuel-air mixture that is provided to the combustion chamber for combustion. Ignition of the fuel-air mixture is accomplished by one or more suitable igniters, and the resulting combustion gases flow along the axial direction A toward and into an annular, first stage turbine nozzle of the HP turbine **132**. The first stage nozzle is defined by an annular flow

channel that includes a plurality of radially-extending, circumferentially-spaced nozzle vanes that turn the gases so that they flow angularly and impinge upon the first stage turbine blades of the HP turbine **132**. The combustion products exit the HP turbine **132** and flow through the LP turbine **134** and exit the core flowpath **142** through the core exhaust nozzle **140** to produce a core air stream or second thrust stream S2. For this embodiment, as noted above, the HP turbine **132** drives the HP compressor **128** via the HP shaft **136** and the LP turbine **134** drives the LP compressor **126**, the primary fan **152**, the mid-fan **190**, and the electric machine **200** via the LP shaft **138**.

(58) The other portion of the second airflow flowing downstream of the mid-fan **190** is split by the splitter **144** into the secondary bypass flowpath **172**. The air enters the secondary bypass flowpath **172** through the secondary bypass inlet **176**. The air flows generally along the axial direction A through the secondary bypass flowpath **172** and is ultimately exhausted from the secondary bypass flowpath **172** through the secondary bypass outlet **178** to produce a third thrust stream S3.

(59) A “third stream” or third thrust stream S3 as used herein means a secondary air stream capable of increasing fluid energy to produce a minority of total propulsion system thrust. In some embodiments, a pressure ratio of the third stream is higher than that of the primary propulsion stream (e.g., a bypass or propeller driven propulsion stream). The thrust may be produced through a dedicated nozzle or through mixing of the secondary air stream with the primary propulsion stream or a core air stream, e.g., into a common nozzle. In certain exemplary embodiments, an operating temperature of the secondary air stream is less than a maximum compressor discharge temperature for the engine. The operating temperature of the third stream may be less than 350 degrees Fahrenheit (such as less than 300 degrees Fahrenheit, such as less than 250 degrees Fahrenheit, such as less than 200 degrees Fahrenheit, and at least as great as an ambient temperature). In certain exemplary embodiments, these operating temperatures may facilitate heat transfer to or from the third stream and a separate fluid stream. Further, in certain exemplary embodiments, the third stream may contribute less than 50% of the total engine thrust (and at least, e.g., 2% of the total engine thrust) at a takeoff condition, or more particularly while operating at a rated takeoff power at sea level, static flight speed, 86 degree Fahrenheit ambient temperature operating conditions. Furthermore in certain exemplary embodiments, aspects of the third stream (e.g., airstream, mixing, or exhaust properties), and thereby the aforementioned exemplary percent contribution to total thrust, may passively adjust during engine operation or be modified purposefully through use of engine control features (such as fuel flow, electric machine power, variable stators, variable inlet guide vanes, valves, variable exhaust geometry, or fluidic features) to adjust or optimize or otherwise improve overall system performance across a broad range of potential operating conditions.

(60) Although three-stream engine **100** has been described and illustrated in FIG. 1 as representing an example three-stream gas turbine engine operable to produce first thrust stream S1, second thrust stream S2, and third thrust stream S3, it will be appreciated that the inventive aspects of the present disclosure may apply to three-stream gas turbine engines having other configurations. For instance, in some embodiments, the three-stream engine **100** may have a centrifugal HP compressor, may not include a booster, may include an electrically-driven booster or LP compressor, etc. Further, in other example embodiments, the primary fan **152** can be ducted by fan casing **157** or outer nacelle, e.g., as shown in FIG. 3A. As shown in FIG. 3A, a bypass passage **159** may be defined between the fan casing **157** and the outer cowl **170**. The first thrust stream S1 may flow through the bypass passage **159**. One or more circumferentially-spaced outlet guide vanes **168** (only one shown in FIG. 3A) can extend between and connect the fan casing **157**, the outer cowl **170**, and the engine core **120** to provide structural support for these components. In such embodiments, the speed reduction gearbox **155** and/or the one or more actuators **158** can be optional. In this regard, the primary fan **152** can be directly coupled with the LP shaft **138**, e.g., in a direct-drive configuration, and/or the primary fan **152** can be a fixed-pitch fan.

(61) In yet other embodiments, the three-stream engine **100** can have other mid-fan and/or splitter

configurations. For instance, as shown in FIGS. 3B and 3C, the nose of the splitter **144** is positioned upstream of the mid-fan **190**. In such embodiments, the mid-fan blades **192** may include an integral splitter which effectively divides the air stream into radially inner and radially outer streams (or second and third streams S2, S3) in proximity to the mid-fan **190** itself. Such a configuration may be termed a blade-on-blade configuration where radially inner and radially outer blades are effectively superimposed upon one another and may be unitarily formed or otherwise fabricated to achieve the split between streams. FIGS. 3B and 3C illustrate embodiments with such a blade-on-blade configuration. Such configurations are described in greater detail in commonly-assigned, issued U.S. Pat. No. 4,043,121, which is incorporated herein by reference.

(62) Further, as shown in FIGS. 3B and 3C, the three-stream engine **100** includes a core array of inlet guide vanes **240A** having a plurality of inlet guide vanes rotatable about their respective central vane axes. The core array of inlet guide vanes **240A** are positioned within the core flowpath **142** upstream of the mid-fan **190**. The pitch of the inlet guide vanes of the core array of inlet guide vanes **240A** can be moved by one or more actuators. Moreover, the three-stream engine **100** also includes a secondary array of inlet guide vanes **240B** having a plurality of inlet guide vanes rotatable about their respective central vane axes. The secondary array of inlet guide vanes **240B** are positioned within the secondary bypass flowpath **172** upstream of the mid-fan **190**. The pitch of the inlet guide vanes of the secondary array of inlet guide vanes **240B** can be moved by one or more actuators. The pitch of the inlet guide vanes of the core array of inlet guide vanes **240A** and the pitch of the inlet guide vanes of the secondary array of inlet guide vanes **240B** can be controlled independently by their respective actuators, e.g., to the same or different pitch angles, or in alternative embodiments, they can be controlled to be pitched in unison.

(63) The core array of inlet guide vanes **240A** can be stacked (i.e., aligned axially) with the secondary array of inlet guide vanes **240B** as shown in FIG. 3B or can be offset from one another along the axial direction A as shown in FIG. 3C. Moreover, in some embodiments as depicted in FIG. 3C, the three-stream engine **100** can include a mid-fan booster **191** that includes a plurality of circumferentially-spaced mid-fan booster blades mechanically coupled with the LP shaft **138**. The mid-fan booster **191** is positioned within the core flowpath **142** upstream of the mid-fan **190** and downstream of the core inlet guide vanes **242A**. The mid-fan booster **191** boosts the core airflow before reaching the mid-fan **190**.

(64) As further shown in FIG. 1 (as well as the embodiments of FIGS. 3A, 3B, 3C), the three-stream engine **100** includes the electric machine **200** mechanically coupled with a rotating component thereof. In this regard, the three-stream engine **100** is an aeronautical hybrid-electric propulsion machine. Particularly, as shown in FIG. 1, the three-stream engine **100** includes the electric machine **200** mechanically coupled with the LP shaft **138**. The electric machine **200** can be directly mechanically connected to the LP shaft **138**, or alternatively, the electric machine **200** can be mechanically coupled with the LP shaft **138** indirectly, e.g., by way of a gearbox **216**. As shown, the electric machine **200** is embedded within the three-stream engine **100** proximate its aft end **116**. Particularly, the electric machine **200** is positioned aft of the mid-fan **190** and at least partially overlapping with or aft of the LP turbine **134** along the axial direction A. Moreover, for this embodiment, the electric machine **200** is positioned inward of the core flowpath **142** along the radial direction R. Although the electric machine **200** is operatively coupled with the LP shaft **138** at an aft end of the LP shaft **138**, the electric machine **200** can be coupled with the LP shaft **138** at any suitable location or can be coupled to other rotating components of the three-stream engine **100**, such as the HP shaft **136**. For instance, in some embodiments, the electric machine **200** can be coupled with the LP shaft **138** and positioned forward of the mid-fan **190** along the axial direction A.

(65) In some embodiments, the electric machine **200** can be an electric motor operable to drive or motor the LP shaft **138**, e.g., when additional thrust is needed. In other embodiments, the electric machine **200** can be an electric generator operable to convert mechanical energy into electrical

energy. In this way, electrical power generated by the electric machine **200** can be directed to various engine and/or aircraft systems. In some embodiments, the electric machine **200** can be a motor/generator with dual functionality.

(66) The electric machine **200** includes a rotor **214** and a stator **224**. The rotor **214** is mechanically coupled with the LP shaft **138** and is rotatable with the LP shaft **138**. The stator **224** can be fixed to a structural support member, such as an aft frame member. The electric machine **200** defines a centerline, which is aligned with or coaxial with the longitudinal axis **112** of the three-stream engine **100** in this example embodiment. The rotor **214** and the stator **224** together define an air gap therebetween. Moreover, the rotor **214** can include a plurality of magnets, such as a plurality of permanent magnets, and the stator **224** can include a plurality of windings or coils. As such, the electric machine **200** may be referred to as a permanent magnet electric machine. However, in other exemplary embodiments, the electric machine **200** may be configured in any suitable manner. For example, the electric machine **200** may be configured as an electromagnetic electric machine, including a plurality of electromagnets and active circuitry, as an induction type electric machine, a switched reluctance type electric machine, a synchronous AC electric machine, an asynchronous electric machine, or as any other suitable electric generator/motor.

(67) The electric machine **200** can be electrically connected to an electric power bus. The electric power bus can be electrically connected to the electric machine **200** at a location inward of the core flowpath **142** along the radial direction **R**. The electric power bus may extend through the core flowpath **142** (e.g., through an aft frame strut) and electrically connect the electric machine **200** to one or more electrical loads (accessory systems, electric/hybrid-electric propulsion devices, etc.), electrical sources (other electric machines, electric energy storage units, etc.), or both. Electrical power can be provided to the electric machine **200** via the electric power bus, e.g., when the electric machine **200** is operating in a drive mode, and electrical power generated by the electric machine **200** can be carried or transmitted to electrical systems via the electric power bus, e.g., when the electric machine **200** is operating in a generator mode.

(68) Although the electric machine **200** has been described and illustrated in FIG. **1** as having a particular configuration, it will be appreciated that the inventive aspects of the present disclosure may apply to electric machines having alternative configurations. For instance, the stator **224** and/or rotor **214** may have different configurations or may be arranged in a different manner than illustrated in FIG. **1**. As one example, in some embodiments, the electric machine **200** may have a tapered configuration in which the rotor **214** and the stator **224** may extend lengthwise along the axial direction **A** at an angle with respect to the longitudinal axis **112**, e.g., so that they are not oriented parallel with the longitudinal axis **112**.

(69) As will be explained further herein, the electric machine **200** can be controlled to effect or control the airflow through the secondary bypass flowpath **172**. In this way, the electric machine **200** can be controlled to optimize or otherwise improve the thrust contribution of the secondary bypass flowpath **172** or to optimize or otherwise improve the thermal management contribution provided by the secondary bypass flowpath **172**, or rather, its interaction with a thermal management system.

(70) As further shown in FIG. **1**, optionally, the three-stream engine **100** can include an electric machine **210** mechanically coupled with the HP shaft **136**. The electric machine **210** can be a motor, a generator, or a combination motor-generator. Like the electric machine **200**, the electric machine **210** can be controlled to effect or control the airflow through the secondary bypass flowpath **172**. In this way, the electric machine **210** can be controlled to optimize or otherwise improve the thrust contribution of the secondary bypass flowpath **172** or to optimize the thermal management contribution provided by the secondary bypass flowpath **172**.

(71) The electric machine **210** can be directly mechanically connected to the HP shaft **136** (as depicted via the solid lines), or alternatively, the electric machine **210** can be mechanically coupled with the HP shaft **138** indirectly, e.g., by way of a gearbox **212** (as shown by the phantom lines).

The directly-coupled electric machine **210** can be embedded within the three-stream engine **100** aft of the LP compressor **126** and forward of the HP compressor **128** along the axial direction A. The directly-coupled electric machine **210** can be positioned inward of the core flowpath **142** along the radial direction R. The indirectly-coupled electric machine **210** and its associated gearbox **212** can be positioned within the core cowl **122** and located outward of the core flowpath **142** along the radial direction R.

(72) FIG. **4** provides a system diagram of a control system **310** for an aircraft **300** according to various embodiments of the present disclosure. The control system **310** can be used to control one or more components of the three-stream engine **100** mounted to the aircraft **300**. While the control system **310** will be described below as being associated with or configured to implement control schemes with respect to the three-stream engine **100** of FIGS. **1** and **2**, it will be appreciated that the example control system **310** of FIG. **4** can be associated with or implement control schemes with respect to other three-stream gas turbine engines as well.

(73) The control system **310** includes, among other things, a supervisory system **320** and an engine controller **340** communicatively coupled thereto. The engine controller **340** can be mounted to the three-stream engine **100** while the supervisory system **320** can be located within a fuselage of the aircraft **300**, for example. The engine controller **340** is communicatively coupled with one or more third stream effectors **400**. Such systems and components can be communicatively coupled in any suitable manner, e.g., via one or more wired or wireless connection links. As will be explained more fully below, the engine controller **340** can control or cause the effector **400** to adjust an airflow through the secondary bypass flowpath **172**. In adjusting the airflow through the secondary bypass flowpath **172**, one or more properties of the airflow can be adjusted, such as the mass flow, temperature, and/or the pressure. By causing the effector **400** to adjust the airflow through the secondary bypass flowpath **172**, the thrust contribution and the thermal management contribution provided by the secondary bypass flowpath **172** can be controlled.

(74) The supervisory system **320** can include one or more processors and one or more memory devices. The one or more processors can perform operations, such as those noted herein. The supervisory system **320** can be configured in a same or similar manner as the computing system **600** provided in FIG. **15**. The supervisory system **320** can be communicatively coupled with various systems of the aircraft **300**, such as a thrust control **322**, one or more electrical systems **324**, and an aircraft thermal management system **326**, denoted as A/C TMS **326** in FIG. **4**. The thrust control **322** can include one or more power or thrust levers positioned within the cockpit or an automated thrust control system, for example. Upon manipulation of the thrust control **322**, a desired amount of thrust may be demanded. The supervisory system **320** can receive signals indicating the demanded thrust in response to manipulation of the thrust control **322**.

(75) The one or more electrical systems **324** can include any aircraft electrical system that draws or generates electrical power. Example electrical systems **324** include, without limitation, air conditioning systems, cockpit displays, cabin accessories, pumps, etc. The aircraft thermal management system **326** can be in thermal communication with one or more of the electrical systems **324**. For instance, the aircraft thermal management system **326** can include one or more heat exchangers, heat sinks, pumps, fluid supply lines, etc. that collectively act to cool one or more components of the electrical systems **324** and potentially other components onboard the aircraft **300**. The supervisory system **320** can receive signals indicated the cooling capacity of the aircraft thermal management system **326** and/or signals indicating electrical power demands from the electrical systems **324**.

(76) As noted, the supervisory system **320** can receive inputs from the thrust control **322**, the electrical systems **324**, and/or the aircraft thermal management system **326**, among other aircraft systems. Based on these inputs, the supervisory system **320** can generate various outputs, such as a thrust demand **330**, denoted as “Thrust DMD(s) **330**” in FIG. **4**, a priority selection **332** indicating a priority of one or more objectives, and an aircraft thermal demand **334**, denoted as “A/C Thermal

DMD(s) 334” in FIG. 4. These generated outputs can be routed to the engine controller 340 as shown in FIG. 4.

(77) The engine controller 340 can include one or more processors and one or more memory devices. The one or more processors can perform operations, such as those noted herein. The engine controller 340 can be configured in a same or similar manner as the computing system 600 provided in FIG. 15. As depicted, the engine controller 340 can include various control modules, including a Power Management module 350, denoted as “PM module 350” in FIG. 4. Upon executing the power management module 350, the one or more processors can output schedule demands, such as an effector schedule 352 and optionally a secondary effector schedule 354, based at least in part on the thrust demand 330, which may be received from the supervisory system 320. The effector schedules 352, 354 can be selected or determined so as to optimize or otherwise improve performance of the three-stream engine 100, for example. In other embodiments, the thrust demand 330 may not be routed through a supervisory system.

(78) Moreover, upon executing the power management module 350, the one or more processors can output one or more thrust limits 356. The one or more thrust limits 356 can indicate a minimum thrust limit, e.g., a minimum thrust required to maintain aircraft controllability. The one or more thrust limits 356 can also indicate a maximum thrust limit, e.g., to prevent the three-stream engine 100 from exceeding its redline speed. The effector schedule 352, 354 and/or the thrust limits 356 can be based on one or more operating parameters in addition to the thrust demand 330. For instance, the effector schedule 352 and/or the thrust limit 356 can be output based on the thrust demand 330 and a temperature at a station of the three-stream engine 100, e.g., a temperature at engine station T2 or at the inlet 176 of the secondary bypass flowpath 172.

(79) The engine controller 340 can also include an engine thermal management system module 360 associated with an engine thermal management system 362, denoted in FIG. 4 as E-TMS module 360 and E-TMS 362, respectively. The engine thermal management system 362 can include one or more heat sinks, heat exchangers, fluid supply lines, pumps, etc. For instance, the engine thermal management system 362 can include the heat exchangers 230 depicted in FIG. 2. In some embodiments, the engine thermal management system 362 is coupled to or integrated with the aircraft thermal management system 326, e.g., as shown in FIG. 4. In other embodiments, the engine thermal management system 362 and the aircraft thermal management system 326 are separate systems. Upon executing the engine thermal management system module 360, the one or more processors can output engine thermal demands 364, denoted as “E-TMS DMD(s) 364” in FIG. 4. The engine thermal demands 364 can indicate that more or less cooling capability is needed, for example.

(80) Further, the engine controller 340 can include an operability module 370. Upon executing the operability module 370, the one or more processors can output one or more operability demands 372, denoted as “OPERABILITY DMD(s) 372” in FIG. 4. The operability module 370 can include various models, tables, etc. relating to operability of the three-stream engine 100 for various operating conditions. Generally, the operability demands 372 can indicate various operating margins limits or lines (e.g., stall margin, surge lines, rotating stall lines) that the three-stream engine 100 must meet in order to maintain operability. The operability demands 372 can be specifically associated with operability of a particular component. In some instances, for example, the operability demands 372 can include an operability demand associated with operability of the mid-fan 190, an operability demand associated with operability of the LP compressor 126, and an operability demand associated with operability of the HP compressor 128.

(81) The engine controller 340 can also include an effector limits module 380. Upon executing the effector limits module 380, the one or more processors can output effector limits 382. The effector limits 382 can indicate the hardware limitations or constraints of the effector 400. For instance, in one example embodiment, the effector 400 can include an array of vanes and an actuator for modulating the position of the vanes. The actuator can define a stroke ranging between a fully

retracted position and a fully extended position. The effector limits can indicate these positions as constraints so that the actuator is not physically moved past its designed limits.

(82) As shown in FIG. 4, the engine controller **340** includes a third stream control module **390**, denoted as “3S CONTROL MODULE **390**” in FIG. 4. The demands **330**, **334**, **364**, **372**, schedules **352**, **354**, limits **356**, **382**, and priority selection **332** can be input into the third stream control module **390**. Particularly, the effector and secondary effector schedules **352**, **354** and thrust limits **356** output from the power management module **350**, the engine thermal demand **364** output from the engine thermal management system module **360**, the operability demands **372** output from the operability module **370**, effector limits **382** output from the effector limits module **380**, and the priority selection **332** and aircraft thermal demands **334** received from the supervisory system **320** can be input into the third stream control module **390**. In executing the third stream control module **390**, the one or more processors can output third stream effector demands **392**, denoted as “3S EFFECTOR DMD(s) **392**” in FIG. 4, based at least in part on one or more of the demands **330**, **334**, **364**, **372**, schedules **352**, **354**, limits **356**, **382**, and priority selection **332**.

(83) The effector demands **392** can be routed to the effector **400**, and in response to the third stream effector demands **392**, the effector **400** can cause adjustment of the airflow through the secondary bypass flowpath **172**. As noted, adjusting the airflow through the secondary bypass flowpath **172** can be an adjustment of one or more properties of the airflow, such as the mass flow, temperature, and/or the pressure. In some embodiments, the secondary effector demands **394** can be routed to the secondary effector **410**, and in response to the secondary effector demands **394**, the secondary effector **410** can assist with adjustment of the airflow through the secondary bypass flowpath **172** to achieve a desired thrust or thermal contribution. Upon causing the effector **400**, and in some instances the secondary effector **410**, to adjust the airflow through the secondary bypass flowpath **172**, one or more engine cycles can be performed, denoted at block **420** in FIG. 4.

(84) One or more sensors **430** of the three-stream engine **100** and/or aircraft **300** can capture one or more operating parameters associated with the three-stream engine **100** and/or aircraft **300**, a current position of the effector **400** and/or secondary effector **410**, and various constraints during the one or more engine cycles performed at block **420**. Sensor data **432** including such captured data can be fed back to the engine controller **340** as shown in FIG. 4. Such sensor data **432** can be utilized by the one or more modules **350**, **360**, **370**, **380**, **390** of the engine controller **340**. For instance, the compressor discharge pressure or pressure at the inlet of the HP compressor **128** can be captured and fed back to the operability module **370** so that, when executed, the operability module **370** can output an operability demand **372** that reflects the actual operating conditions within the three-stream engine **100**. Moreover, the temperature of the airflow flowing through the secondary bypass flowpath **172** can be captured and fed to the engine thermal management system module **360** so that, when executed, the engine thermal management system module **360** can output an engine thermal demand **364** that reflects the actual operating conditions within the secondary bypass flowpath **172**, and accordingly, can demand more or less cooling to achieve a desired thermal target. The sensor data **432** can also include other feedback, such as the rotational speed of one or more of the shafts **136**, **138** (potentially corrected by an engine temperature at a particular engine station), an electrical characteristic associated with the electric machine **200** (e.g., a voltage or electric current associated with the electric machine **200**), etc., as well as other sensor inputs that may facilitate deriving certain parameters, such as an engine pressure ratio and/or an exhaust gas temperature.

(85) As further shown in FIG. 4, the engine controller **340** can provide feedback data **440** to the supervisory system **320**. Specifically, thrust feedback, thermal feedback, and electric feedback, among other possible feedback, can be provided to the supervisory system **320** by the engine controller **340**. The thrust feedback can indicate a predicted thrust output of the three-stream engine **100**. In some instances, the thrust feedback can also specifically indicate the thrust output from the secondary bypass flowpath **172**. The thermal feedback can indicate a cooling capacity of one or

more heat sinks associated with the engine thermal management system **362**, for example. The electric feedback can indicate one or more characteristics associated with the electrical systems or components of the three-stream engine **100**. For instance, the voltage or electric current associated with the electric machine **200** can be fed back to the supervisory system **320**. The supervisory system **320** can utilize the feedback data **440** in generating demands, e.g., for a subsequent timestep of the control system **310**.

(86) With the control system **310** described generally, an example manner in which the one or more processors of the engine controller **340** can cause the effector **400** to adjust the airflow through the secondary bypass flowpath **172** will now be provided. For this example embodiment, the effector **400** is the array of inlet guide vanes **240** positioned upstream of the mid-fan **190**. In causing the array of inlet guide vanes **240** to adjust the airflow through the secondary bypass flowpath **172**, the one or more processors of the engine controller **340** are configured to cause adjustment of a pitch of the inlet guide vanes **240** about their respective central vane axes **242**. For instance, based on the third-stream effector demands **392** output from the engine controller **340**, the actuator **244** can collectively pitch the inlet guide vanes **240** about their respective central vane axes **242**.

Consequently, the airflow through the secondary bypass flowpath **172** is adjusted. That is, at least one of the mass flow, temperature, and pressure of the airflow through the secondary bypass flowpath **172** is adjusted by moving the position of the inlet guide vanes **240**.

(87) With reference now to FIGS. **1**, **2**, **4**, and **5**, the one or more processors of the engine controller **340** can cause the array of inlet guide vanes **240** to adjust the airflow through the secondary bypass flowpath **172** in accordance with the demands **330**, **334**, **364**, **372**, schedules **352**, **354**, limits **356**, **382**, and priority selection **332** provided to the third stream control module **390**. Particularly, FIG. **5** graphically depicts inlet guide vane position scheduled as a function of a power level of the three-stream engine **100**. An effector schedule **352** associated with the inlet guide vane position is depicted as a function of power level. The effector schedule **352** can be generated and output by execution of the power management module **350** as depicted in FIG. **4**. For this embodiment, the effector schedule **352** has a negative exponential shape as depicted in FIG. **5**. The effector schedule **352** is a default or base schedule that the effector demands **392** are generated in accordance with unless the effector schedule **352** is constrained by another demand or limit.

(88) In accordance with the effector schedule **352**, the lower the power level of the three-stream engine **100**, the more closed the inlet guide vanes **240** are moved. Moving the inlet guide vanes **240** more closed decreases the mass flow of the airflow through the secondary bypass flowpath **172** and also cools the temperature of the airflow therethrough. Lower power levels are associated with higher propulsive efficiency of the three-stream engine **100** as indicated in FIG. **5**. In contrast, in accordance with the effector schedule **352**, the higher the power level of the three-stream engine **100**, the more open the inlet guide vanes **240** are moved. Moving the inlet guide vanes **240** to a more open position increases the mass flow of the airflow through the secondary bypass flowpath **172** and also increases the temperature of the airflow therethrough. Higher power levels are associated with higher specific thrust of the three-stream engine **100** as also indicated in FIG. **5**.

(89) As further shown in FIG. **5**, in addition to the effector schedule **352**, various demands and limits are depicted. Particularly, the operability demand **372**, a thermal demand **368**, the thrust limit **356**, and the effector limits **382** are depicted. The thermal demand **368**, operability demand **372**, and thrust limit **356** are movable along the y-axis of the graph as represented by the double-sided arrows positioned adjacent the operability demand **372**, the thermal demand **368**, and the thrust limit **356**. Specifically, the thermal demand **368** is movable along the y-axis of the graph based on inputs from the aircraft thermal management system **326** and/or the engine thermal management system **362**. The thermal demand **368** can be a summation of the engine thermal demand **364** and the aircraft thermal demand **334**, for example. If additional cooling is demanded, for instance, the thermal demand **368** can be moved upward along the y-axis. This may, for example, increase the cooling capacity of the heat exchangers **230** positioned in the secondary bypass flowpath **172**. In

contrast, if less cooling is demanded, the thermal demand **368** can be moved downward along the y-axis of the graph.

(90) The operability demand **372** is movable along the y-axis of the graph based on the sensor data **432** indicating one or more operating parameters associated with the three-stream engine **100**, such as the rate of change of the compressor discharge pressure or pressure at the inlet of the HP compressor **128**. The thrust limit **356** is movable along the y-axis of the graph based on the sensor data **432** indicating one or more operating parameters associated with the three-stream engine **100**, such as the temperature or pressure at a station of the engine. The effector limits **382** are fixed constraints in this example embodiment. The effector limits **382** include an open limit and a closed limit in this example.

(91) In some embodiments, the position of the inlet guide vanes **240** is selected in accordance with the effector schedule **352** unless other demands and/or limits intersect with or are selected as being of higher priority, which may cause the position of the inlet guide vanes **240** to deviate from the effector schedule **352**. The priority of the demands and/or limits may be preselected, e.g., by the supervisory system **320** and/or the engine controller **340**. Examples are provided below.

(92) EXAMPLE 1: As a first example, with reference to FIG. 6 in addition to FIGS. 1, 2, 4, and 5, FIG. 6 provides a graph depicting inlet guide vane position as a function of time. In this example, the power level of the three-stream engine **100** remains constant, and thus, the effector schedule **352** is shown as a constant function. Further, for this example, the supervisory system **320** and/or engine controller **340** seeks to prioritize the cooling of various components of the aircraft **300**. Accordingly, at time **t1**, the thermal demand **368** begins to increase, which represents a demand for increased cooling capacity. Also, before and at time **t1**, the position of the inlet guide vanes **240** is set in accordance with the effector schedule **352**, as represented by the effector demand **392** tracing over the effector schedule **352** prior to and at time **t1**. The effector demand **392** is shown in dashed lines in FIG. 6.

(93) At time **t2**, the thermal demand **368** intersects with the effector schedule **352** and continues to increase thereafter. The intersection of the thermal demand **368** with the effector schedule **352** indicates that the thermal demand **368** is of higher priority than the effector schedule **352**. Notably, at time **t2**, the effector demand **392** begins to trace along the thermal demand **368** rather than the effector schedule **352**. In accordance with the effector demands **392**, the inlet guide vanes **240** are moved or pitched about their respective central vanes axes **242**. In this instance, to meet the increase in thermal demand, the inlet guide vanes **240** are pitched to a more closed position, which reduces the temperature and mass flow of the airflow through the secondary bypass flowpath **172**. Advantageously, this may increase the rate of thermal exchange between the heat exchangers **230** and the airflow through the secondary bypass flowpath **172**. The inlet guide vanes **240** are gradually pitched more closed from time **t2** to time **t3**.

(94) At time **t3**, the thermal demand **368** and effector demand **392** tracing along the thermal demand **368** intersect with the thrust limit **356**. To avoid passing the thrust limit **356**, at **t3**, the effector demand **392** begins to trace along the thrust limit **356** even though the thermal demand **358** continues to increase after time **t3**. As shown, the effector demand **392** and consequently the position of the inlet guide vanes **240** is held for a time at the thrust limit **356** (until time **t5**) to provide the coolest possible airflow through secondary bypass flowpath **172** whilst not affecting the ability of the three-stream engine **100** to maintain aircraft controllability. While the thrust contribution from the secondary bypass flowpath **172** is sacrificed somewhat when the inlet guide vanes **240** are moved more closed, the thermal contribution that the secondary bypass flowpath **172** provides to one or more thermal management systems can be increased.

(95) At time **t4**, the increased thermal demand for one or more of the thermal management systems of the aircraft **300** is no longer needed. Thus, at **t4**, the thermal demand **368** starts to decrease. The thermal demand **368** can be decreased linearly as shown in FIG. 6. At time **t5**, the thermal demand **368** intersects with the thrust limit **356** and the effector demand **392** tracing along the thrust limit

356. As less cooling is needed, at time **t5**, the effector demand **392** begins to trace along the thermal demand **368** rather than the thrust limit **356**. In accordance with the effector demands **392**, the inlet guide vanes **240** are moved or pitched about their respective axes **242**. In this instance, to meet the decrease in thermal demand **368** and to increase the thrust contribution provided by the secondary bypass flowpath **172**, the inlet guide vanes **240** are pitched to a more open position, which increases the temperature and mass flow of the airflow through the secondary bypass flowpath **172**. The inlet guide vanes **240** are gradually pitched more open from time **t5** to time **t6**.

(96) At time **t6**, the thermal demand **368** intersects once again with the effector schedule **352**. Thus, the effector demand **392** begins to trace along the effector schedule **352** rather than the thermal demand **368** at time **t6**. This allows for optimization or improvement of the thrust contribution provided by the secondary bypass flowpath **172** whilst still providing a nominal thermal contribution. In accordance with the effector demands **392**, the inlet guide vanes **240** are no longer moved more open but rather are held in position at time **t6**.

(97) EXAMPLE 2: As a second example, with reference to FIG. 7 in addition to FIGS. 1, 2, 4, and 5, FIG. 7 provides a graph depicting inlet guide vane position as a function of time as well as a thrust demand **330** as a function of time. In this example, the power level of the three-stream engine **100** does not remain constant. Rather, the three-stream engine **100** is subjected to an engine transient. Particularly, as depicted in FIG. 7, the thrust demand **330** increases at time **t1**. This causes the effector schedule **352** to decrease along the y-axis at time **t1**. As the effector demand **392** traces along the effector schedule **352**, at time **t1**, the inlet guide vanes **240** are moved more open to increase the mass flow and temperature of the air flowing through the secondary bypass flowpath **172**, thereby increasing the thrust contribution provided by the secondary bypass flowpath **172** to the net propulsive thrust. As a result of the increase thrust demanded at time **t1**, at time **t2**, the operability demand **372** begins to increase along the y-axis, e.g., as the compressor operating line moves closer to the surge line or stall line. The thrust demand **330** increases to time **t4**.

(98) At time **t3**, the operability demand **372** intersects with the effector schedule **352**. To prevent inoperability of the three-stream engine **100**, the effector demand **392** is determined as being of higher priority and begins to trace along the operability demand **372** instead of the effector schedule **352**. Consequently, at time **t3**, in accordance with the effector demand **392**, the inlet guide vanes **240** cease moving to a more open position and move more closed to decrease the mass flow of the air flowing through the secondary bypass flowpath **172**. This may ensure operability of the three-stream engine **100**.

(99) At time **t4**, the thrust demand **330** is no longer increasing. Thus, the operability demand **372** ceases to increase along the y-axis of the graph. The thrust demand **330** is held constant from time **t4** to time **t5**. At time **t5**, a decrease in thrust is demanded as represented by the decreasing thrust demand **330**. When the thrust demand **330** decreases at time **t5**, the operability demand **372** begins to decrease along the y-axis of the graph, e.g., as the compressor operating line moves away from the surge line or stall line.

(100) At time **t6**, the operability demand **372** intersects with the effector schedule **352** once again. Thus, at time **t6**, the effector demand **392** begins to trace along the effector schedule **352** rather than the operability demand **372**. As less thrust is demanded, at time **t6**, in accordance with the effector demand **392**, the inlet guide vanes **240** are moved to a more closed position, which decreases the mass flow and temperature of the air flowing through the secondary bypass flowpath **172**. At time **t7**, the effector schedule **352** ceases increasing along the y-axis as the thrust demand **330** ceases decreasing.

(101) EXAMPLE 3: As a third example, with reference to FIG. 8 in addition to FIGS. 1, 2, 4, and 5, FIG. 8 provides a graph depicting inlet guide vane position as a function of time as well as a thrust demand **330** as a function of time. In this example, increased cooling is demanded and the power level of the three-stream engine **100** is subjected to an engine transient. As depicted in FIG. 8, at time **t1**, an increased amount of cooling is demanded as reflected by the thermal demand **368**

increasing along the y-axis of the graph. Prior to and at time **t1**, the effector demand **392** traces along the effector schedule **352**. At time **t2**, the increasing thermal demand **368** intersects with the effector schedule **352**. Accordingly, the effector demand **392** begins to trace along the thermal demand **368** rather than the effector schedule **352** at time **t2**. In accordance with the effector demand **392**, the inlet guide vanes **240** are moved to a more closed position, which cools down the air flowing through the secondary bypass flowpath **172** thereby increasing the cooling capacity of the heat exchangers **230**.

(102) At time **t3**, the thrust demand **330** increases, e.g., in response to a pilot manipulating the thrust control **322**. This causes the effector schedule **352** to decrease along the y-axis at time **t3**. In this example, the supervisory system **320** has made a priority selection **332** indicating that meeting the thrust demand **330** during this transient operation is of higher priority than meeting the thermal demand **368**. Accordingly, the effector demand **392** aborts tracing along the thermal demand **368** and returns to follow the effector schedule **352** as safely and quickly as possible. In accordance with the effector demand **392**, from time **t3** to time **t4** the inlet guide vanes **240** are moved to a more open position, which increases the mass flow and temperature of the air flowing through the secondary bypass flowpath **172**, which increases the thrust contribution provided by the secondary bypass flowpath **172**. As a result of the increase thrust demanded at time **t3**, the operability demand **372** begins to increase along the y-axis, e.g., as the compressor operating line moves closer to the surge line or stall line. The thrust demand **330** increases to time **t5**.

(103) At time **t4**, as the effector demand **392** traces along the effector schedule **352**, the operability demand **372** intersects with the effector schedule **352**. To prevent inoperability of the three-stream engine **100**, the effector demand **392** begins to trace along the operability demand **372** instead of the effector schedule **352**. Consequently, at time **t4**, in accordance with the effector demand **392**, the inlet guide vanes **240** cease moving to a more open position and move more closed to decrease the mass flow of the air flowing through the secondary bypass flowpath **172**.

(104) At time **t5**, the thrust demand **330** is no longer increasing. Thus, the operability demand **372** ceases to increase along the y-axis of the graph. The thrust demand **330** is held constant from time **t5** to time **t6**. At time **t6**, a decrease in thrust is demanded as represented by the decreasing thrust demand **330**. When the thrust demand **330** decreases at time **t6**, the operability demand **372** begins to decrease along the y-axis of the graph, e.g., as the compressor operating line moves away from the surge line or stall line.

(105) At time **t7**, the operability demand **372** intersects with the effector schedule **352** once again. Thus, at time **t7**, the effector demand **392** begins to trace along the effector schedule **352** rather than the operability demand **372**. As less thrust is demanded, at time **t7**, in accordance with the effector demand **392**, the inlet guide vanes **240** are moved to a more closed position, which decreases the mass flow and temperature of the air flowing through the secondary bypass flowpath **172**.

(106) At time **t8**, the thrust demand **330** ceases to decrease and levels off. The effector schedule **352** correspondingly ceases to increase along the y-axis of the graph and levels off. As the engine transient operation has completed at time **t8** (as reflected by the thrust demand **330**) and the thermal demand **368** has still not crossed or re-intersected with the effector schedule **352**, the engine controller **340** seeks to meet the thermal demand **368**. Thus, the effector demand **392** deviates from the effector schedule **352** and increases along the y-axis to meet the thermal demand **368** at time **t8**. At time **t9**, the effector demand **392** reaches the thrust limit **356** before reaching the thermal demand **368**. Accordingly, the effector demand **392** traces along the thrust limit **356** instead of the thermal demand **368** so as not to negatively affect the controllability of the aircraft **300**.

(107) After time **t9** and before time **t10**, less thermal demand **368** is requested, and thus the thermal demand **368** begins to decrease along the y-axis of the graph. At time **t10**, the thermal demand **368** intersects with the thrust limit **356**. Accordingly, the effector demand **392** begins to trace along the thermal demand **368** at time **t10**. The thermal demand **368** continues to decrease after time **t10**. In accordance with the effector demand **392**, this causes the inlet guide vanes **240** to move to a more

open position. At time **t11**, the thermal demand **368** intersects with the effector schedule **352**. As a result, the effector demand **392** begins to trace along the effector schedule **352**.

(108) EXAMPLE 1, EXAMPLE 2, and EXAMPLE 3 provide example manners in which the engine controller **340** can control the effector **400**, or array of inlet guide vanes **240** in these example embodiments, to cause adjustment of the airflow through the secondary bypass flowpath **172**. Particularly, the one or more processors of the engine controller **340** are configured to: determine an effector demand **392** based at least in part on an interaction between: i) the effector schedule **352** determined based at least in part on the thrust demand **330**, the thrust demand **330** being associated with thrust to be produced by the three-stream engine **100**; ii) the thermal demand **368** associated with a thermal management system coupled to or integrated with the three-stream engine **100**; iii) the operability demand **372** associated with operability of the three-stream engine **100**; and iv) the thrust limit **356** associated with controllability of the aircraft **300**. Accordingly, in such embodiments, the one or more processors are configured to cause the effector **400** to adjust an airflow through the secondary bypass flowpath **172** based at least in part on the effector demand **392**. The effector demand **392** can be determined in accordance with the effector schedule **352** unless one of the demands or limits is selected or determined as being of higher priority, for example.

(109) Stated another way, the one or more processors of the engine controller **340** are configured to: determine an effector demand **392** based at least in part on an effector schedule **352** and one or more constraints, the effector schedule **352** being determined based at least in part on the thrust demand **330** associated with thrust to be produced by the three-stream engine **100**, the one or more constraints including at least one of: i) the thermal demand **368** associated with a thermal management system coupled to or integrated with the three-stream engine **100**; ii) an operability demand **372** associated with operability of the three-stream engine **100**; and iii) the thrust limit **356** associated with controllability of the aircraft **300**. The one or more processors of the engine controller **340** are configured to cause the effector **400** to adjust an airflow through the secondary bypass flowpath **172** based at least in part on the effector demand **392**.

(110) In accordance with the inventive aspects of the present disclosure, the effector **400** can be or can include other components in addition or alternatively to the inlet guide vanes **240**. For instance, in some embodiments, the effector **400** can be the variable nozzle **250** positioned along the secondary bypass flowpath **172**. In such embodiments, in causing the variable nozzle **250** to adjust the airflow through the secondary bypass flowpath **172**, the one or more processors of the engine controller **340** are configured to cause the variable nozzle **250** to change position. For instance, on one hand, when more cooling is needed, the one or more processors of the engine controller **340** can be configured to cause the variable nozzle **250** to move to a more open position. On the other hand, when additional thrust is needed, the one or more processors of the engine controller **340** can be configured to cause the variable nozzle **250** to move to a more closed position. As noted, in FIG. 2, the variable nozzle **250** is shown in the fully open position via solid lines and in the fully closed position via the phantom lines. The variable nozzle **250** can be moved by the actuator **252** to vary the exit area through the secondary bypass outlet **178** in accordance with the effector demand **392** output by the engine controller **340**.

(111) FIG. 9 provides a graph depicting variable nozzle position scheduled as a function of a power level of the three-stream engine of FIGS. 1 and 2. As depicted, the shape of the effector schedule **352** associated with the variable nozzle position is similar to the shape of the effector schedule **352** associated with the inlet guide vanes depicted in FIG. 5. However, on the y-axis of the graph of FIG. 9, the fully closed position is the minimum position while the fully open position is the maximum position. In contrast, for the inlet guide vane position depicted in FIG. 5, the fully open position is the minimum position while the fully closed position is the maximum position. In this regard, the open and closed positions of the inlet guide vanes and the variable nozzle as they relate to the maximum and minimums are reversed.

(112) As further shown in FIG. 9, the effector limits **382**, thrust limit **356**, operability demand **372**, and thermal demand **368** are graphically represented. In view of the teachings provided herein and the noted differences between how the variable nozzle **250** is to be controlled to provide increased cooling capability or an increased thrust contribution compared to how the inlet guide vanes **240** are controlled to achieve the same, it will be appreciated that the effector demand **392** for the variable nozzle **250** can be generated by the engine controller **340** and the variable nozzle **250** can be controlled in accordance with the effector demand **392** in a same or similar manner described above with respect to control of the inlet guide vanes **240**.

(113) In some further embodiments, the effector **400** can include the array of inlet guide vanes **240** positioned upstream of the mid-fan **190** and the variable nozzle **250** positioned along the secondary bypass flowpath **172**. In this regard, the engine controller **340** can output effector demands **392** for controlling the inlet guide vanes **240** and effector demands **392** for controlling the variable nozzle **250**.

(114) In yet other embodiments, in addition or alternatively to the effector **400** being or including the inlet guide vanes **240**, and/or the variable nozzle **250**, the effector **400** is or can include the primary fan **152**. In such embodiments, in causing the effector **400**, or the primary fan **152** in this example, to adjust the airflow through the secondary bypass flowpath **172**, the one or more processors of the engine controller **340** are configured to cause at least one of: i) adjustment of a pitch of fan blades **154** of the primary fan **152**; and ii) adjustment of a rotational speed of the primary fan **152**. In some embodiments, both the pitch of fan blades **154** of the primary fan **152** and the speed of the rotational speed of the primary fan **152** can be adjusted to effect a change in thrust or thermal contribution provided by the secondary bypass flowpath **172**.

(115) The one or more processors of the engine controller **340** can cause the one or more actuators **158** to pitch the fan blades **154** about their respective central blade axes **156** to a more open or closed position to effect a change in thrust or thermal contribution provided by the secondary bypass flowpath **172**. Moreover, the one or more processors of the engine controller **340** can cause more or less fuel to be provided to the combustor **130**, which effectively changes the rotational speed of the LP shaft **138** and consequently the primary fan **152** mechanically coupled thereto.

(116) FIG. 10 provides a contour graph depicting various fan speed lines arranged by pitch as a function of a corrected thrust of the three-stream engine of FIGS. 1 and 2. The pitch of the primary fan blades **154** is represented on the y-axis of the graph of FIG. 10 while the corrected thrust is represented on the x-axis of the graph. There are three example speed contour lines shown, including a first speed line SP1, a second speed line SP2, and a third speed line SP3. The first speed line SP1 represents a higher speed than the second speed line SP2, and the second speed line SP2 represents a higher speed than the third speed line SP3. More or less than three speed lines are contemplated. The speed lines SP1, SP2, SP3 effectively depict various speed/pitch combinations that may achieve a given corrected thrust. It will be appreciated that the contour graph of FIG. 10 may alternatively be presented as various pitch lines arranged by fan speed as a function of a corrected thrust of the three-stream engine of FIGS. 1 and 2.

(117) As further shown in FIG. 10, the effector limits **382**, thrust limit **356**, operability demand **372**, and thermal demand **368** are graphically represented. The thrust limit **356** is movable along the x-axis of the graph of FIG. 10. In addition, for this embodiment, a torque limit **384** and a speed avoidance limit **386** are graphically represented as well. The torque limit **384** and the speed avoidance limit **386** can be output by the effector limits module **380**, for example. The torque limit **384** can indicate a maximum torque at which the primary fan **152** may operate before exceeding its designed limit. The speed avoidance limit **386** can indicate a speed at which the primary fan **152** negatively affects the dynamic response of the LP system and/or HP system of the three stream engine **100**. For instance, the speed avoidance limit **386** can indicate a speed at which rub events are predicted to occur (a rub event may include the rubbing of a turbine blade on a shroud, for example), a speed at which vibration damage associated with the LP shaft **138** and/or HP shaft **136**

are predicted to occur, and/or a speed at which mechanical and/or thermal fatigue is unacceptable. The torque limit **384** and the speed avoidance limit **386** are movable along the y-axis of the graph of FIG. **10** as denoted by the double-sided arrows positioned adjacent thereto.

(118) In view of the teachings provided herein, it will be appreciated that the primary fan **152** can be controlled to provide increased cooling capability or an increased thrust contribution based at least in part on the effector demand **392** (FIG. **4**). Particularly, the primary fan **152** can be controlled in accordance with the effector demand **392**, e.g., in a same or similar manner described above with respect to control of the inlet guide vanes **240**.

(119) In other embodiments, in addition or alternatively to the effector **400** being or including the inlet guide vanes **240**, and/or the variable nozzle **250**, and/or the primary fan **152**, the effector **400** is or can include the electric machine **200** mechanically coupled with the LP shaft **138** and/or the electric machine **210** mechanically coupled with the HP shaft **136**. In such example embodiments, in causing the effector **400**, or the electric machine **200** and/or electric machine **210** in this example, to adjust the airflow through the secondary bypass flowpath **172**, the one or more processors of the engine controller **340** are configured to cause the electric machine **200** to adjust a torque applied to the LP shaft **138** and/or the electric machine **210** to adjust a torque applied to the HP shaft **136**. Adjusting the torque applied by the electric machine **200** on the LP shaft **138** affects the rotational speed of the mid-fan **190**, which is also mechanically coupled with the LP shaft **138**. Adjusting the torque applied by the electric machine **210** on the HP shaft **136** affects the rotational speed of the HP shaft **136**.

(120) In some instances, when the electric machine **200** acts as a motor, the one or more processors of the engine controller **340** can cause a drive associated with the electric machine **200** to drive the LP shaft **138** with more or less torque. When more thrust is demanded, for example, the one or more processors can command the drive to cause the electric machine **200** to increase the torque applied on the LP shaft **138**. This effectively causes the rotational speed of the mid-fan **190** to increase, causing the properties of the airflow through the secondary bypass flowpath **172** to change, and consequently, the thrust contribution provided by the secondary bypass flowpath **172** to increase. When less thrust is demanded, the one or more processors can command the drive to cause the electric machine **200** to decrease the torque applied on the LP shaft **138**. This causes the rotational speed of the mid-fan **190** to decrease, causing the properties of the airflow through the secondary bypass flowpath **172** to change, and as a result, the thrust contribution provided by the secondary bypass flowpath **172** decreases. The thermal contribution provided by the secondary bypass flowpath **172** can be modulated as well by changing the torque applied by the electric machine **200** on the LP shaft **138**.

(121) In other instances, such as during cruise operation, the electric machine **200** may act as a generator. In such instances, the one or more processors can cause the electric machine **200** to operate more or less efficiently. This, in effect, causes the electric machine **200** to adjust a torque applied to the LP shaft **138**, and consequently, the thrust and/or thermal contribution provided by the secondary bypass flowpath **172**.

(122) In some instances, the one or more processors of the engine controller **340** can cause one or both of the electric machine **200** and the electric machine **210** to apply more or less torque on or to their respective shafts **138**, **186**, which ultimately affects a Power Ratio defined as LP Power/Total Power, wherein the LP Power is the electrical power output by the electric machine **200** and the Total Power is the total electrical power output by the electric machine **200** and the electric machine **210**.

(123) FIG. **11** provides a graph depicting various effector schedules provided as a power ratio as a function of a power level of the three-stream engine of FIGS. **1** and **2**. Particularly, FIG. **11** provides a first effector schedule **352-1** associated with a first flight condition (e.g., a first altitude) and a second effector schedule **352-2** associated with a second flight condition (e.g., a second altitude) that is different than the first flight condition. More or less than two effector schedules are

contemplated. For instance, an effector schedule can be provided for each contemplated flight condition. The effector schedule to be utilized in controlling one or both of the electric machines **200**, **210** can be based on the present flight conditions. The effector schedule having a flight condition that most closely matches the present flight condition can be selected, for example.

(124) For the first effector schedule **352-1**, for power ratios over 1.0, the electric machine **200** is controlled to generate electrical power and the electric machine **210** is controlled to motor or drive the HP shaft **136**. At a power ratio of 1.0, the electric machine **200** is controlled to continue generating electrical power and the electric machine **210** is controlled to idle. For power ratios below 1.0 and above 0.0, both electric machines **200**, **210** are controlled to generate electrical power. At a power ratio of 0.0, the electric machine **210** is controlled to continue generating electrical power and the electric machine **200** is controlled to idle. For power ratios below 0.0, the electric machine **210** is controlled to generate electrical power and the electric machine **200** is controlled to motor or drive the LP shaft **138**.

(125) For the second effector schedule **352-2**, for power ratios over 1.0, the electric machine **200** is controlled to generate electrical power and the electric machine **210** is controlled to motor or drive the HP shaft **136**. However, at power level PL1 and higher, the slope of the second effector schedule **352-2** flattens out or has a slope of zero, which causes the electric machine **210** to idle, while the electric machine **200** is controlled to continue generating electrical power.

(126) As further shown in FIG. **11**, the operability demand **372** and thermal demands are graphically represented. For this embodiment, two thermal demands are depicted, including a first thermal demand **368-1** and a second thermal demand **368-2**. The first thermal demand **368-1** and the second thermal demand **368-2** are movable along the y-axis of the graph of FIG. **11** as denoted by the double-sided arrows positioned adjacent thereto. The first thermal demand **368-1** indicates a thermal demand associated with adjusting a power split between the electric machine **200** and the electric machine **210** based on a capability of the secondary bypass flowpath **172** to reject heat produced by the electrical systems associated with the three-stream engine **100** (including the electric machine **200** and/or the electric machine **210**) and/or the aircraft **300** (FIG. **4**). Essentially, the first thermal demand **368-1** indicates a demand to adjust the power split of the electric machines **200**, **210** to adjust the heat produced by the electric machines **200**, **210**, or rather by adjusting the “hot side”. This can be accomplished by looking at the capability of the secondary bypass flowpath **172** (or the capability of heat exchangers therein) to reject heat and also looking at the temperature of the electric machines, their associated thermal busses, etc.

(127) The second thermal demand **368-2** indicates a thermal demand associated with adjusting a power split between the electric machine **200** and the electric machine **210** to influence one or more properties of the airflow through the secondary bypass flowpath **172** to ultimately raise or lower the cooling capability of the secondary bypass flowpath **172**. Essentially, the second thermal demand **368-2** indicates a demand to adjust the power split of the electric machines **200**, **210** to influence the thermal management capabilities of the secondary bypass flowpath **172**, or rather to influence or adjust the “cold side”. This can be accomplished by looking at the temperature of the electric machines **200**, **210**, their associated thermal busses, etc.

(128) In view of the teachings provided herein, it will be appreciated that the electric machine **200** and/or electric machine **210** can be controlled to provide increased cooling capability or an increased thrust contribution or a change in the heat generated thereby based at least in part on the effector demand **392** (FIG. **4**). Particularly, the electric machine **200** and/or electric machine **210** can be controlled in accordance with the effector demand **392**, e.g., in a same or similar manner described above with respect to control of the inlet guide vanes **240**. That is, the effector schedule having a flight condition that most closely matches the present flight condition can be selected and the effector demand can be generated in accordance with the selected effector schedule unless otherwise constrained, e.g., by one of the thermal demands **368-1**, **368-2**, the operability demand **372**, or some other constraint noted herein but not depicted in FIG. **11**.

(129) In some further embodiments, optionally, the three-stream engine **100** can include the secondary effector **410** positioned downstream of the mid-fan **190** along the core flowpath **142**. The secondary effector **410** can be controlled to assist or facilitate adjustment of the airflow through the secondary bypass flowpath **172** in conjunction with one or more of the disclosed effectors **400**. In such embodiments, as will be explained further below, the one or more processors of the engine controller **340** are configured to cause the secondary effector **410** to assist with adjustment of the airflow through the secondary bypass flowpath **172** based at least in part on the secondary effector demand **394**.

(130) For instance, in some example embodiments, the three-stream engine **100** can include a compressor having one or more stages of compressors rotor blades and compressor stator vanes, wherein at least one stage of compressor stator vanes are variable stator vanes. For instance, as shown in FIGS. **1** and **2**, the LP compressor **126** includes one or more stages of compressors rotor blades and compressor stator vanes, and notably, at least one stage of the compressor stator vanes are variable compressor stator vanes. Particularly, the variable compressor stator vanes are the variable booster inlet guide vanes **260**. For this embodiment, the secondary effector can be or can include the booster inlet guide vanes **260**.

(131) With reference now to FIG. **12** in addition to FIGS. **1-2** and **3-4**, FIG. **12** provides a graph depicting booster variable stator vane position scheduled as a function of a power level of the three-stream engine **100**. As depicted, the shape of the secondary effector schedule **354** associated with the booster inlet guide vanes **260** is similar to the shape of the effector schedule **352** depicted in FIG. **5** associated with the inlet guide vanes **240**.

(132) The secondary effector schedule **354** associated with a position of the booster inlet guide vanes **260** is depicted as a function of power level. The secondary effector schedule **354** can be generated and output by execution of the power management module **350** as depicted in FIG. **4**. For this embodiment, the secondary effector schedule **354** has a negative exponential shape as depicted in FIG. **12**. The secondary effector schedule **354** is a default or base schedule that the secondary effector demands **394** are generated in accordance with unless the secondary effector schedule **354** is constrained by another demand or limit.

(133) In accordance with the secondary effector schedule **354**, the lower the power level of the three-stream engine **100**, the more closed the booster inlet guide vanes **260** are moved. Moving the booster inlet guide vanes **260** more closed back pressures the mid-fan **190**, thereby affecting the airflow through the secondary bypass flowpath **172**. In contrast, in accordance with the secondary effector schedule **354**, the higher the power level of the three-stream engine **100**, the more open the booster inlet guide vanes **260** are moved. Moving the booster inlet guide vanes **260** more open increases the pumping ability of the mid-fan **190**, which changes one or more properties of the airflow through the secondary bypass flowpath **172**.

(134) As further shown in FIG. **12**, in addition to the secondary effector schedule **354**, various demands and limits are depicted. Particularly, an operability demand **372A** associated with operability of the secondary effector **410** (the booster inlet guide vanes **260** in this example), an operability demand **372B** associated with operability of the mid-fan **190**, and effector limits **382** are depicted. The operability demands **372A**, **372B** are movable along the y-axis of the graph. The operability demand **372A** is movable along the y-axis of the graph based on the sensor data **432** indicating one or more operating parameters associated with the LP compressor **126**. The operability demand **372B** is movable along the y-axis of the graph based on the sensor data **432** indicating one or more operating parameters associated with the mid-fan **190**.

(135) In some embodiments, the one or more processors of the engine controller **340** are configured to determine the secondary effector demand **394** based at least in part on the interaction between: i) the secondary effector schedule **354** determined based at least in part on the thrust demand **330**, ii) the operability demand **372A** associated with operability of the secondary effector **410** (or in this example the booster inlet guide vanes **260**); and iii) the operability demand **372B** associated with

operability of the mid-fan **190**. As the operability demand **372B** associated with operability of the mid-fan **190** is considered, the control logic of the secondary effector **410** can be said to be ganged with the control logic of the mid-fan **190**.

(136) Generally, the position of the booster inlet guide vanes **260** is selected in accordance with the secondary effector schedule **354** unless other demands and/or limits **372A**, **372B**, **382** intersect with or are selected as being of higher priority, which may cause the position of the booster inlet guide vanes **260** to deviate from the secondary effector schedule **354**. The priority of the demands and/or limits may be preselected, e.g., by the supervisory system **320** and/or the engine controller **340**. In some embodiments, the operability demand **372A** associated with the booster inlet guide vanes **260** is selected as being of higher priority than the operability demand **372B** associated with the mid-fan **190**.

(137) Stated another way, the one or more processors of the engine controller **340** are configured to determine the secondary effector demand **394** based at least in part on a secondary effector schedule **354** and one or more secondary constraints, the secondary effector schedule **354** being determined based at least in part on the thrust demand **330**, the one or more secondary constraints including at least one of: i) an operability demand **372A** associated with operability of the secondary effector **410**; and ii) an operability demand associated with operability of the mid-fan **190**. Further, the one or more processors of the engine controller **340** are configured to cause the secondary effector **410** to assist with adjustment of the airflow through the secondary bypass flowpath **172** based at least in part on the secondary effector demand **394**.

(138) In yet other embodiments, in addition or alternatively to the booster inlet guide vanes **260**, the secondary effector **410** can include the variable bleed valve **270**. In such embodiments, in causing the secondary effector **410**, or the variable bleed valve **270** in this example, to assist with adjustment of the airflow through the secondary bypass flowpath **172**, the one or more processors of the engine controller **340** are configured to cause adjustment of a position of the variable bleed valve **270**. The one or more processors of the engine controller **340** can cause adjustment of the position of the variable bleed valve **270** so that core air from the core flowpath **142** is directed into one of the offboard bleed duct **272** and the secondary bleed duct **274**.

(139) With reference now to FIG. **13** in addition to FIGS. **1-2** and **3-4**, FIG. **13** provides a graph depicting variable bleed valve position scheduled as a function of a power level of the three-stream engine **100**. The secondary effector schedule **354** associated with a position of the variable bleed valve **270** is depicted as a function of power level. The shape of the secondary effector schedule **354** associated with the variable bleed valve **270** is generally linear having a negative slope and transitions to a constant function at higher power levels. The secondary effector schedule **354** can be generated and output by execution of the power management module **350** as depicted in FIG. **4**. The secondary effector schedule **354** is a default or base schedule that the secondary effector demands **394** are generated in accordance with unless other demands or limits are of higher priority than the secondary effector schedule **354**.

(140) In accordance with the secondary effector schedule **354**, the lower the power level of the three-stream engine **100**, the more open the variable bleed valve **270** is moved. In contrast, in accordance with the secondary effector schedule **354**, the higher the power level of the three-stream engine **100**, the more closed the variable bleed valve **270** is moved.

(141) When air is bled through the variable bleed valve **270** to the secondary bypass flowpath **172** via the secondary bleed duct **274**, moving the variable bleed valve **270** more open back pressures the mid-fan **190**, which changes one or more properties of the airflow through the secondary bypass flowpath **172**. In addition, in such a bleed arrangement, moving the variable bleed valve **270** more closed increases the pumping ability of the mid-fan **190**, which changes one or more properties of the airflow through the secondary bypass flowpath **172**.

(142) When air is bled through the variable bleed valve **270** offboard via the offboard bleed duct **272**, moving the variable bleed valve **270** more open reduces the pressure on the mid-fan **190**,

which changes one or more properties of the airflow through the secondary bypass flowpath **172**. Further, in such a bleed arrangement, moving the variable bleed valve **270** more closed increases the pressure on the mid-fan **190**, which changes one or more properties of the airflow through the secondary bypass flowpath **172**.

(143) As further shown in FIG. **13**, in addition to the secondary effector schedule **354**, various demands and limits are depicted. Particularly, an operability demand **372A** associated with operability of the secondary effector **410** (the variable bleed valve **270** in this example), an operability demand **372B** associated with operability of the mid-fan **190**, an extraction demand **374** indicating a minimum open position the variable bleed valve is to be set (e.g., to expel debris or to be opened during takeoff and climb operations), and effector limits **382** are depicted. The operability demands **372A**, **372B** are movable along the y-axis of the graph. The operability demand **372A** is movable along the y-axis of the graph based on the sensor data **432** indicating one or more operating parameters associated with the variable bleed valve **270**. The operability demand **372B** is movable along the y-axis of the graph based on the sensor data **432** indicating one or more operating parameters associated with the mid-fan **190**. The extraction demand **374** can be output by the operability module **370** and is also movable along the y-axis of the graph. As noted, the extraction demand **374** indicates a minimum open position the variable bleed valve **270** is to be set. For instance, during takeoff and climb operations, the variable bleed valve **270** may be set more open to prevent engine stalls. However, during cruise operations, the variable bleed valve **270** may be able to be more closed as operation is typically more steady state.

(144) In some embodiments, the one or more processors of the engine controller **340** are configured to determine the secondary effector demand **394** for the variable bleed valve **270** based at least in part on the interaction between i) the secondary effector schedule **354** determined based at least in part on the thrust demand **330**, ii) the operability demand **372A** associated with operability of the secondary effector **410** (or in this example the variable bleed valve **270**); iii) the operability demand **372B** associated with operability of the mid-fan **190**; and iv) the extraction demand **374** indicating the minimum open position the variable bleed valve **270** is to be set. As noted above, as the operability demand **372B** associated with operability of the mid-fan **190** is considered, the control logic of the secondary effector **410** can be said to be ganged with the control logic of the mid-fan **190**.

(145) Generally, the position of the variable bleed valve **270** is selected in accordance with the secondary effector schedule **354** unless other demands and/or limits **372A**, **372B**, **374**, **382** intersect with or are selected as being of higher priority, which may cause the position of the variable bleed valve **270** to deviate from the secondary effector schedule **354**. The priority of the demands and/or limits may be preselected, e.g., by the supervisory system **320** and/or the engine controller **340**. In some embodiments, the operability demand **372A** associated with the variable bleed valve **270** is selected as being of higher priority than the operability demand **372B** associated with the mid-fan **190**, which is selected as being of higher priority than the extraction demand **374**.

(146) Stated differently, in some embodiments, the one or more processors of the engine controller **340** are configured to determine the secondary effector demand **394** for the variable bleed valve **270** based at least in part on the secondary effector schedule **354** and the one or more secondary constraints, the one or more secondary constraints including at least one of: i) the operability demand **372A** associated with operability of the secondary effector **410**; ii) the operability demand **372B** associated with operability of the mid-fan **190**; and iii) an extraction demand **374** indicating a minimum open position the variable bleed valve **270** is to be set. Moreover, the one or more processors of the engine controller **340** are configured to cause adjustment of the position of the variable bleed valve **270** so that core air from the core flowpath **142** is directed into one of the offboard bleed duct **272** and the secondary bleed duct **274**.

(147) FIG. **14** provides a flow diagram for a method **500** of operating a three-stream engine for an aircraft according to one example embodiment of the present disclosure. For instance, the method

500 can be used to operate the three-stream engine **100** of FIGS. **1** and **2**, FIG. **3**, as well as other three-stream engines. It will be appreciated that method **500** is discussed herein to describe exemplary aspects of the present subject matter and is not intended to be limiting.

(148) At **502**, the method **500** includes determining, by one or more processors, an effector demand for an effector of the three-stream gas turbine engine, the three-stream engine having a primary fan and a mid-fan and defining a primary bypass flowpath, a secondary bypass flowpath, and a core flowpath, the effector demand being determined based at least in part on an interaction between: i) an effector schedule determined based at least in part on a thrust demand, the thrust demand being associated with thrust to be produced by the three-stream gas turbine engine; ii) a thermal demand associated with a thermal management system coupled to or integrated with the three-stream gas turbine engine; iii) an operability demand associated with operability of the three-stream gas turbine engine; and iv) a thrust limit associated with controllability of the aircraft. In some implementations, for example, the effector demand can be determined in accordance with the effector schedule. In this regard, the effector schedule can be a base schedule. However, when one of the demands or limits is selected or determined as being of a higher priority, the effector demand can be determined in accordance with that constraint, which may be a demand or limit.

(149) In some implementations, the effector is an array of inlet guide vanes positioned upstream of a mid-fan of the three-stream engine. The mid-fan is positioned downstream of a primary fan, which may be ducted or unducted. In such implementations, in causing the effector to adjust the airflow through the secondary bypass flowpath, the method can include causing, by the one or more processors, adjustment of a position of the inlet guide vanes, e.g., to a more open or closed position.

(150) In other implementations, the effector is a variable nozzle positioned along the secondary bypass flowpath. In such implementations, in causing the effector to adjust the airflow through the secondary bypass flowpath, the method can include causing, by the one or more processors, adjustment of a position of the variable nozzle, e.g., to a more open or closed position.

(151) In yet other implementations, the effector is an electric machine mechanically coupled with the shaft to which the mid-fan is also mechanically coupled. In such implementations, in causing the effector to adjust the airflow through the secondary bypass flowpath, the method can include causing, by the one or more processors, the electric machine to adjust a torque applied to the shaft.

(152) In some further implementations, the effector is a primary fan positioned upstream of the mid-fan. In such implementations, in causing the effector to adjust the airflow through the secondary bypass flowpath, the method can include causing, by the one or more processors, at least one of: i) adjustment of a pitch of fan blades of the primary fan; and ii) adjustment of a rotational speed of the primary fan.

(153) In yet other implementations, the effector can include the inlet guide vanes positioned upstream of the mid-fan, the variable nozzle, the electric machine, and the primary fan, or any possible combination thereof.

(154) In some implementations, the method **500** includes, determining, by the one or more processors, the effector schedule based at least in part on the thrust demand and data indicating one or more operating parameters associated with the three-stream gas turbine engine. The method **500** can further include outputting, by the one or more processors, the effector demand in accordance with the effector schedule, e.g., as shown up to time **t1** and after time **t6** in FIG. **6**.

(155) In some implementations, the method **500** includes, determining, by the one or more processors, the operability demand based at least in part on the data indicating the one or more operating parameters associated with the three-stream gas turbine engine. The method **500** can further include determining, by the one or more processors, that the operability demand is a highest priority constraint. In response to the operability demand being determined as the highest priority constraint, the effector demand is output in accordance with the operability demand, e.g., as shown at time **t3** to time **t6** in FIG. **7**.

(156) In some implementations, the method **500** includes, determining, by the one or more processors, the thrust limit based at least in part on the thrust demand. The method **500** can also include determining, by the one or more processors, that the thrust limit is a highest priority constraint. In response to the thrust limit being determined as the highest priority constraint, the effector demand is output in accordance with the thrust limit, e.g., as shown at time **t3** to time **t5** in FIG. **6**.

(157) In some implementations, the method **500** includes, receiving, by the one or more processors, the thermal demand. The thermal demand received can be associated with an aircraft thermal management system, an engine thermal management system, or a summation of the two. The method **500** can also include determining, by the one or more processors, that the thermal demand is a highest priority constraint. In response to the thermal demand being determined as the highest priority constraint, the effector demand is output in accordance with the thermal demand, e.g., as shown at time **t2** to time **t3** in FIG. **6** as well as time **t5** to time **t6** in FIG. **6**.

(158) At **504**, the method **500** includes causing, by the one or more processors, the effector to adjust an airflow through a secondary bypass flowpath defined by the three-stream engine based at least in part on the effector demand.

(159) In some further implementations, optionally, the method can include determining, by the one or more processors, a secondary effector demand associated with a second effector positioned downstream of the mid-fan along the core flowpath based at least in part on an interaction between: i) a secondary effector schedule determined based at least in part on the thrust demand; ii) an operability demand associated with operability of the secondary effector; and iii) an operability demand associated with operability of the mid-fan. Further, in such implementations, the method can include causing, by the one or more processors, the secondary effector to assist with adjustment of the airflow through the secondary bypass flowpath based at least in part on the secondary effector demand.

(160) In some implementations, the three-stream gas turbine engine has a compressor having one or more stages of compressors rotor blades and compressor stator vanes, wherein at least one stage of compressor stator vanes are variable stator vanes. For instance, the variable stator vanes can be booster inlet guide vanes of a low pressure compressor or booster of the engine. In such implementations, the secondary effector can be the variable stator vanes. In causing the secondary effector to assist with adjustment of the airflow through the secondary bypass flowpath, the one or more processors can be configured to cause adjustment of a position of the variable stator vanes.

(161) In yet other implementations, the three-stream gas turbine engine has a low pressure compressor positioned along the core flowpath downstream of the mid-fan, a high pressure compressor positioned along the core flowpath downstream of the low pressure compressor, and a variable bleed valve positioned downstream of the low pressure compressor and upstream of the high pressure compressor. In such implementations, the secondary effector can be the variable bleed valve. In causing the secondary effector to assist with adjustment of the airflow through the secondary bypass flowpath, the one or more processors can be configured to cause adjustment of a position of the variable bleed valve. Particularly, in some implementations, the method can include determining, by the one or more processors, the secondary effector demand based at least in part on an interaction between: i) the secondary effector schedule determined based at least in part on the thrust demand; ii) the operability demand associated with operability of the secondary effector; and iii) the operability demand associated with operability of the mid-fan; and iv) an extraction demand indicating a minimum open position the variable bleed valve is to be set. In such implementations, the one or more processors are configured to cause adjustment of the position of the variable bleed valve so that core air from the core flowpath is directed into one of the offboard bleed duct and the secondary bleed duct.

(162) In some implementations, the three-stream gas turbine engine defines a primary fan radius to mid-fan radius ratio as being equal to or greater than 2.0 and less than or equal to 6.5, the primary

fan radius to mid-fan radius ratio being defined by a radius spanning between a longitudinal axis defined by the three-stream gas turbine engine and a leading edge tip of one of the primary fan blades of the primary fan to a radius spanning between the longitudinal axis and a leading edge tip of one of the mid-fan blades of the mid-fan.

(163) FIG. 15 provides a block diagram of an example computing system 600. The computing system 600 can be used to implement the aspects disclosed herein. The computing system 600 can include one or more computing device(s) 602. The engine controller 340 and the supervisory system 320 disclosed herein can be constructed and may operate in a same or similar manner as one of the computing devices 602, for example.

(164) As shown in FIG. 15, the one or more computing device(s) 602 can each include one or more processor(s) 604 and one or more memory device(s) 606. The one or more processor(s) 604 can include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, or other suitable processing device. The one or more memory device(s) 606 can include one or more computer-readable media, including, but not limited to, non-transitory computer-readable medium or media, RAM, ROM, hard drives, flash drives, and other memory devices, such as one or more buffer devices.

(165) The one or more memory device(s) 606 can store information accessible by the one or more processor(s) 604, including computer-readable or computer-executable instructions 608 that can be executed by the one or more processor(s) 604. The instructions 608 can be any set of instructions or control logic that when executed by the one or more processor(s) 604, cause the one or more processor(s) 604 to perform operations. The instructions 608 can be software written in any suitable programming language or can be implemented in hardware. In some embodiments, the instructions 608 can be executed by the one or more processor(s) 604 to cause the one or more processor(s) 604 to perform operations.

(166) The memory device(s) 606 can further store data 610 that can be accessed by the processor(s) 604. For example, the data 610 can include sensor data such as engine parameters, model data, logic data, etc., as described herein. The data 610 can include one or more table(s), function(s), algorithm(s), model(s), equation(s), etc. according to example embodiments of the present disclosure.

(167) The one or more computing device(s) 602 can also include a communication interface 612 used to communicate, for example, with the other components of the aircraft. The communication interface 612 can include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, or other suitable components.

(168) The technology discussed herein makes reference to computer-based systems and actions taken by and information sent to and from computer-based systems. It will be appreciated that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein can be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications can be implemented on a single system or distributed across multiple systems.

(169) Although specific features of various embodiments may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the present disclosure, any feature of a drawing may be referenced and/or claimed in combination with any feature of any other drawing.

(170) This written description uses examples to disclose the subject matter, including the best mode, and also to enable any person skilled in the art to practice the subject matter, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the subject matter is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the

claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

(171) To summarize, three-stream engines architecturally arranged and operable to implement one or more of the disclosed control schemes may allow for optimized or otherwise improved performance and constraint handling of such three-stream engines. One or more effectors, and optionally one or more secondary effectors, can be controlled to adjust a thrust contribution provided by a secondary bypass flowpath to a net propulsive thrust of a three-stream engine as well as a thermal contribution provided by the secondary bypass flowpath to an associated thermal management system. Competing demands, limits, and priorities can be considered in controlling the one or more effectors to balance the thrust and thermal demands of the three-stream engine, as well as the operability and hardware limitations of the three-stream engine. The controllability of the aircraft or vehicle to which the three-stream engine is mounted may also be considered.

(172) Further aspects of the present disclosure are provided by the subject matter of the following clauses:

(173) 1. A three-stream gas turbine engine for an aircraft, comprising: a shaft; a primary fan mechanically coupled with the shaft; a mid-fan positioned downstream of the primary fan and mechanically coupled with the shaft; an engine core; a core cowl surrounding at least a portion of the engine core, a core flowpath being defined between the engine core and the core cowl; an outer cowl surrounding at least a portion of the core cowl, a secondary bypass flowpath being defined between the core cowl and the outer cowl; an effector; and one or more processors configured to: determine an effector demand based at least in part on an effector schedule and one or more constraints, the effector schedule being determined based at least in part on a thrust demand associated with thrust to be produced by the three-stream gas turbine engine, the one or more constraints including at least one of: i) a thermal demand associated with a thermal management system coupled to or integrated with the three-stream gas turbine engine; ii) an operability demand associated with operability of the three-stream gas turbine engine; and iii) a thrust limit associated with controllability of the aircraft; and cause the effector to adjust an airflow through the secondary bypass flowpath based at least in part on the effector demand.

(174) 2. The three-stream gas turbine engine of any preceding clause, wherein the effector is an array of inlet guide vanes positioned upstream of the mid-fan.

(175) 3. The three-stream gas turbine engine of any preceding clause, wherein the effector is a variable nozzle positioned along the secondary bypass flowpath.

(176) 4. The three-stream gas turbine engine of any preceding clause, wherein the effector includes an array of inlet guide vanes positioned upstream of the mid-fan and a variable nozzle positioned along the secondary bypass flowpath.

(177) 5. The three-stream gas turbine engine of any preceding clause, wherein the effector is an electric machine mechanically coupled with the shaft, and wherein in causing the effector to adjust the airflow through the secondary bypass flowpath, the one or more processors are configured to: cause the electric machine to adjust a torque applied to the shaft.

(178) 6. The three-stream gas turbine engine of any preceding clause, wherein the effector is the primary fan, and wherein in causing the effector to adjust the airflow through the secondary bypass flowpath, the one or more processors are configured to: cause at least one of: i) adjustment of a pitch of fan blades of the primary fan; and ii) adjustment of a rotational speed of the primary fan.

(179) 7. The three-stream gas turbine engine of any preceding clause, further comprising: a secondary effector positioned downstream of the mid-fan along the core flowpath, and wherein the one or more processors are configured to: determine a secondary effector demand based at least in part on a secondary effector schedule and one or more secondary constraints, the secondary effector schedule being determined based at least in part on the thrust demand, the one or more secondary constraints including at least one of: i) an operability demand associated with operability of the

secondary effector; and ii) an operability demand associated with operability of the mid-fan; and cause the secondary effector to assist with adjustment of the airflow through the secondary bypass flowpath based at least in part on the secondary effector demand.

(180) 8. The three-stream gas turbine engine of any preceding clause, further comprising: a compressor having one or more stages of compressors rotor blades and compressor stator vanes, wherein at least one stage of compressor stator vanes are variable stator vanes, and wherein the secondary effector is the variable stator vanes.

(181) 9. The three-stream gas turbine engine of any preceding clause, further comprising: a low pressure compressor positioned along the core flowpath downstream of the mid-fan; a high pressure compressor positioned along the core flowpath downstream of the low pressure compressor; and a variable bleed valve positioned downstream of the low pressure compressor and upstream of the high pressure compressor, and wherein the secondary effector is the variable bleed valve, and wherein in causing the secondary effector to assist with adjustment of the airflow through the secondary bypass flowpath, the one or more processors are configured to: cause adjustment of a position of the variable bleed valve.

(182) 10. The three-stream gas turbine engine of any preceding clause, wherein the three-stream gas turbine engine defines at least one of an offboard bleed duct and a secondary bleed duct, the offboard bleed duct providing fluid communication between the core flowpath and offboard the three-stream gas turbine engine, the secondary bleed duct providing fluid communication between the core flowpath and the secondary bypass flowpath, and wherein the one or more processors are configured to: determine the secondary effector demand based at least in part on the secondary effector schedule and the one or more secondary constraints, the one or more secondary constraints including at least one of: i) the operability demand associated with operability of the secondary effector; ii) the operability demand associated with operability of the mid-fan; and iii) an extraction demand indicating a minimum open position the variable bleed valve is to be set, and wherein the one or more processors are configured to cause adjustment of the position of the variable bleed valve so that core air from the core flowpath is directed into one of the offboard bleed duct and the secondary bleed duct.

(183) 11. The three-stream gas turbine engine of any preceding clause, wherein the one or more processors are configured to: determine the effector schedule based at least in part on the thrust demand and data indicating one or more operating parameters associated with the three-stream gas turbine engine; and output the effector demand in accordance with the effector schedule.

(184) 12. The three-stream gas turbine engine of any preceding clause, wherein the one or more processors are configured to: determine the operability demand based at least in part on data indicating one or more operating parameters associated with the three-stream gas turbine engine; and determine that the operability demand is a highest priority constraint, and wherein, in response to the operability demand being the highest priority constraint, the effector demand is output in accordance with the operability demand.

(185) 13. The three-stream gas turbine engine of any preceding clause, wherein the one or more processors are configured to: determine the thrust limit based at least in part on the thrust demand; and determine that the thrust limit is a highest priority constraint, and wherein, in response to the thrust limit being the highest priority constraint, the effector demand is output in accordance with the thrust limit.

(186) 14. The three-stream gas turbine engine of any preceding clause, wherein the one or more processors are configured to: receive the thermal demand, and determine that the thermal demand is a highest priority constraint, and wherein, in response to the thermal demand being the highest priority constraint, the effector demand is output in accordance with the thermal demand.

(187) 15. The three-stream gas turbine engine of any preceding clause, wherein the three-stream gas turbine engine defines a primary fan radius to mid-fan radius ratio as being equal to or greater than 2.0 and less than or equal to 6.5, the primary fan radius to mid-fan radius ratio being defined

by a radius spanning between a longitudinal axis defined by the three-stream gas turbine engine and a leading edge tip of one primary fan blade of the primary fan to a radius spanning between the longitudinal axis and a leading edge tip of one mid-fan blade of the mid-fan.

(188) 16. A non-transitory computer readable medium comprising computer-executable instructions, which, when executed by one or more processors associated with a three-stream gas turbine engine having a primary fan and a mid-fan and defining a core flowpath, a primary bypass flowpath, and a secondary bypass flowpath, cause the one or more processors to: determine an effector demand associated with an effector of the three-stream gas turbine engine, the effector demand being determined based at least in part on an interaction between: i) an effector schedule determined based at least in part on a thrust demand, the thrust demand being associated with thrust to be produced by the three-stream gas turbine engine; ii) a thermal demand associated with a thermal management system coupled to or integrated with the three-stream gas turbine engine; iii) an operability demand associated with operability of the three-stream gas turbine engine; and iv) a thrust limit associated with controllability provided by the three-stream gas turbine engine; and cause the effector to adjust an airflow through the secondary bypass flowpath based at least in part on the effector demand.

(189) 17. The non-transitory computer readable medium of any preceding clause, wherein the effector includes at least one of an array of inlet guide vanes positioned upstream of the mid-fan, a variable nozzle positioned along the secondary bypass flowpath.

(190) 18. The non-transitory computer readable medium of any preceding clause, wherein the effector includes at least one of electric machine mechanically coupled with a shaft to which the mid-fan is mechanically coupled and the primary fan.

(191) 19. The non-transitory computer readable medium of any preceding clause, wherein, when the computer-executable instructions are executed, the one or more processors are configured to: determine a secondary effector demand associated with a secondary effector positioned downstream of the mid-fan along the core flowpath, the one or more processors determine the secondary effector demand by considering: i) a secondary effector schedule determined based at least in part on the thrust demand; ii) an operability demand associated with operability of the secondary effector; and iii) an operability demand associated with operability of the mid-fan; and cause the secondary effector to assist with adjustment of the airflow through the secondary bypass flowpath based at least in part on the secondary effector demand.

(192) 20. An aircraft, comprising: a thermal management system; a three-stream gas turbine engine defining a core flowpath, a primary bypass flowpath, and a secondary bypass flowpath, the thermal management system being coupled to or integrated with the three-stream gas turbine engine, the three-stream gas turbine engine comprising: a shaft; a primary fan mechanically coupled with the shaft; a mid-fan positioned downstream of the primary fan and mechanically coupled with the shaft; an effector; and one or more processors configured to: determine an effector demand based at least in part on a thrust demand being associated with thrust to be produced by the three-stream gas turbine engine and a thermal demand associated with the thermal management system; and cause the effector to adjust an airflow through the secondary bypass flowpath based at least in part on the effector demand.

(193) 21. A method of operating a three-stream gas turbine engine for an aircraft, the method comprising: determining, by one or more processors, an effector demand for an effector of the three-stream gas turbine engine, the three-stream engine having a primary fan and a mid-fan and defining a primary bypass flowpath, a secondary bypass flowpath, and a core flowpath, the effector demand being determined based at least in part on an interaction between i) an effector schedule determined based at least in part on a thrust demand, the thrust demand being associated with thrust to be produced by the three-stream gas turbine engine; ii) a thermal demand associated with a thermal management system coupled to or integrated with the three-stream gas turbine engine; iii) an operability demand associated with operability of the three-stream gas turbine engine; and iv) a

thrust limit associated with controllability of the aircraft; and causing, by the one or more processors, the effector to adjust an airflow through a secondary bypass flowpath defined by the three-stream engine based at least in part on the effector demand.

Claims

1. A three-stream gas turbine engine for an aircraft, comprising: a shaft; a primary fan mechanically coupled with the shaft; a mid-fan positioned downstream of the primary fan and mechanically coupled with the shaft; an engine core; a core cowl surrounding at least a portion of the engine core, a core flowpath being defined between the engine core and the core cowl; an outer cowl surrounding at least a portion of the core cowl, a secondary bypass flowpath being defined between the core cowl and the outer cowl; an effector, wherein the effector is an array of inlet guide vanes positioned upstream of the mid-fan; a thermal management system having one or more heat exchangers disposed in the secondary bypass flowpath; and one or more processors configured to: determine an effector demand based on an effector schedule and a thermal demand associated with the thermal management system; and cause the effector to adjust an airflow through the secondary bypass flowpath based at least in part on the effector demand, wherein causing the effector to adjust the airflow through the secondary bypass flowpath based at least in part on the effector demand includes: setting the array of inlet guide vanes to a first position in accordance with the effector schedule when the thermal demand is at a first level; and adjusting the array of inlet guide vanes to a second position when the thermal demand is at a second level that is greater than the first level, the second position being more closed than the first position in order to reduce temperature and mass flow of airflow through the secondary bypass flowpath and increase a rate of thermal exchange between the one or more heat exchangers and the airflow.
2. The three-stream gas turbine engine of claim 1, further comprising: a secondary effector positioned along the core flowpath; and a compressor having one or more stages of compressors rotor blades and compressor stator vanes, wherein at least one stage of compressor stator vanes are variable stator vanes, and wherein the secondary effector is the variable stator vanes.
3. The three-stream gas turbine engine of claim 1, further comprising: a low pressure compressor positioned along the core flowpath downstream of the mid-fan; a high pressure compressor positioned along the core flowpath downstream of the low pressure compressor; and a variable bleed valve positioned downstream of the low pressure compressor and upstream of the high pressure compressor, and wherein a secondary effector is the variable bleed valve, and wherein in causing the secondary effector to assist with adjustment of the airflow through the secondary bypass flowpath, the one or more processors are configured to: cause adjustment of a position of the variable bleed valve.
4. The three-stream gas turbine engine of claim 1, wherein the three-stream gas turbine engine defines a primary fan radius to mid-fan radius ratio as being equal to or greater than 2.0 and less than or equal to 6.5, the primary fan radius to mid-fan radius ratio being defined by a radius spanning between a longitudinal axis defined by the three-stream gas turbine engine and a leading edge tip of one primary fan blade of the primary fan to a radius spanning between the longitudinal axis and a leading edge tip of one mid-fan blade of the mid-fan.
5. The three-stream gas turbine engine of claim 1, wherein the one or more processors are further configured to: receive a temperature at the one or more heat exchangers from an associated temperature sensor operable to sense the temperature at the one or more heat exchangers; and cause the effector to adjust an airflow through the secondary bypass flowpath based at least in part on the temperature at the one or more heat exchangers.
6. The three-stream gas turbine engine of claim 1, wherein causing the effector to adjust the airflow through the secondary bypass flowpath based at least in part on the effector demand includes: holding a position of the array of inlet guide vanes for a period of time to avoid exceeding a thrust

limit.

7. A non-transitory computer readable medium comprising computer-executable instructions, which, when executed by one or more processors associated with a three-stream gas turbine engine having a primary fan and a mid-fan and defining a core flowpath, a primary bypass flowpath, and a secondary bypass flowpath, cause the one or more processors to: determine an effector demand associated with an effector of the three-stream gas turbine engine based on an effector schedule and a thermal demand associated with a thermal management system having one or more heat exchangers disposed in the secondary bypass flowpath, wherein the effector is an array of inlet guide vanes positioned upstream of the mid-fan; and cause the effector to adjust an airflow through the secondary bypass flowpath based at least in part on the effector demand, wherein causing the effector to adjust the airflow through the secondary bypass flowpath based at least in part on the effector demand includes: setting the array of inlet guide vanes to a first position in accordance with the effector schedule when the thermal demand is at a first level; and adjusting the array of inlet guide vanes to a second position when the thermal demand is at a second level that is greater than the first level, the second position being more closed than the first position in order to reduce temperature and mass flow of airflow through the secondary bypass flowpath and increase a rate of thermal exchange at the thermal management system.

8. An aircraft, comprising: a three-stream gas turbine engine defining a core flowpath, a primary bypass flowpath, and a secondary bypass flowpath; and a thermal management system having one or more heat exchangers disposed in the secondary bypass flowpath, the three-stream gas turbine engine comprising: a shaft; a primary fan mechanically coupled with the shaft; a mid-fan positioned downstream of the primary fan and mechanically coupled with the shaft; an engine core; a core cowl surrounding at least a portion of the engine core, the core flowpath being defined between the engine core and the core cowl; an outer cowl surrounding at least a portion of the core cowl, the secondary bypass flowpath being defined between the core cowl and the outer cowl; an effector, wherein the effector is an array of inlet guide vanes positioned upstream of the mid-fan; and one or more processors configured to: determine an effector demand based on an effector schedule and a thermal demand associated with the thermal management system; and cause the effector to adjust an airflow through the secondary bypass flowpath based at least in part on the effector demand, wherein causing the effector to adjust the airflow through the secondary bypass flowpath based at least in part on the effector demand includes: setting the array of inlet guide vanes to a first position in accordance with the effector schedule when the thermal demand is at a first level; and adjusting the array of inlet guide vanes to a second position when the thermal demand is at a second level that is greater than the first level, the second position being more closed than the first position in order to reduce temperature and mass flow of airflow through the secondary bypass flowpath and increase a rate of thermal exchange between the one or more heat exchangers and the airflow.

9. The aircraft of claim 8, wherein the three-stream gas turbine engine further comprises: a secondary effector positioned along the core flowpath; and a compressor having one or more stages of compressors rotor blades and compressor stator vanes, wherein at least one stage of compressor stator vanes are variable stator vanes, and wherein the secondary effector is the variable stator vanes.
